# HSC Mathematics Extension 1 & 2

Proof by Mathematical Induction

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*Download PDF and resources:*

`https://github.com/vuhung16au/math-olympiad-ml`

*Read online at Vu's Maths Hub:*

`https://vumaths.com/`

*"Mathematics is the language with which God has written the universe."*

Galileo Galilei

# Contents



 

# 1 Introduction

## 1.1 Project Overview

This booklet compiles high quality mathematical induction problems curated specifically for the HSC Mathematics Extension 2 syllabus. Every task can be attacked with induction (sometimes combined with algebraic manipulation, inequalities, recurrence relations, or integrals). Detailed reasoning showcases common techniques such as weak and strong induction, structural induction on algebraic forms, and inductive proofs for inequalities.

## 1.2 Target Audience

The explanations are crafted for Extension 2 students aiming to deepen their proof-writing skills. Each solution explicitly states the induction hypothesis, justification of the base case, and reasoning for the inductive step so that high-school learners can follow every transition.

## 1.3 How to Use This Booklet

- Read the overview and induction primer before attempting the problems.
- Attempt problems in Part 1 without hints; compare against the detailed solutions to understand model reasoning.
- Use the upside-down hints in Part 2 only after making a genuine attempt.
- Revisit problems after a few days and try to re-derive the arguments without notes to reinforce technique.

## 1.4 Induction Primer

Mathematical induction is a two-step method:

1. **Base Case:** Verify the statement for the initial value (often $n = 1$ or $n = 0$), ensuring the proposition is anchored.
2. **Inductive Step:** Assume the statement is true for $n = k$ (or all values up to $k$ for strong induction) and show it holds for $n = k + 1$. Clearly state the hypothesis and highlight algebraic transformations used to bridge from $k$ to $k + 1$.

We will reference these two steps in every solution so students can trace the logic effortlessly.

# 2 Part 1: Problems and Solutions (Detailed)

Part 1 contains three sets of problems—basic, medium, and advanced. Each set provides five problems. For every problem we present a comprehensive induction-based solution without any hints so that learners focus on the full reasoning trail.

 

## 2.1 Basic Induction Problems

**Problem 2.1: Telescoping Harmonic Sum**

Show that for every integer $n \geq 1$,

$$\sum_{k=1}^{n} \frac{1}{k(k+1)} = \frac{n}{n+1}.$$

**Solution 2.1**

**Base case** $(n = 1)$**.** We have $\frac{1}{1\cdot 2} = \frac{1}{2}$ and $\frac{1}{1+1} = \frac{1}{2}$, so the statement is true for $n = 1$.
**Inductive step.** Assume the statement holds for some $n = k$, i.e.

$$\sum_{r=1}^{k} \frac{1}{r(r+1)} = \frac{k}{k+1}.$$

Then

$$\begin{aligned} \sum_{r=1}^{k+1} \frac{1}{r(r+1)} &= \underbrace{\sum_{r=1}^{k} \frac{1}{r(r+1)}}_{\frac{k}{k+1}} + \frac{1}{(k+1)(k+2)} \\ &= \frac{k}{k+1} + \frac{1}{(k+1)(k+2)} \\ &= \frac{k(k+2)+1}{(k+1)(k+2)} = \frac{k^2+2k+1}{(k+1)(k+2)} \\ &= \frac{(k+1)^2}{(k+1)(k+2)} = \frac{k+1}{k+2}. \end{aligned}$$

Therefore the identity is true for $n = k + 1$, so by induction it holds for all $n \geq 1$.

**Takeaways 2.1**

Partial fractions $\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$ produce telescoping sums,
so the algebra in the inductive step stays simple and transparent.

**Problem 2.2: Divisibility of $n^3 + 2n$**

Prove that $n^3 + 2n$ is divisible by 12 for every even integer $n$.

**Solution 2.2**

Let $P(n)$ be the statement for the even integer $n$. Write $n = 2m$ for some integer $m$.
**Base case** $(n = 2)$**.** $2^3 + 2 \cdot 2 = 8 + 4 = 12$, which is divisible by 12.
**Inductive step.** Assume $P(2m)$ holds, so $2m$ even implies $(2m)^3 + 2(2m) = 8m^3 + 4m$ is a multiple of 12. Consider the next even integer $2(m + 1)$:

$$\begin{aligned} (2m+2)^3 + 2(2m+2) &= 8(m+1)^3 + 4(m+1) \\ &= 8(m^3 + 3m^2 + 3m + 1) + 4m + 4 \\ &= (8m^3 + 4m) + 24m^2 + 24m + 12. \end{aligned}$$

By the hypothesis, $8m^3 + 4m$ is divisible by 12, and each remaining term has factor 12. Hence the whole expression is divisible by 12, completing the induction.

### Takeaways 2.2

Factoring $n$ as $2m$ keeps arithmetic simple and highlights the repeated factor of 12.

### Problem 2.3: Nine divides $7^n + 2^n$

Show that $7^n + 2^n$ is divisible by 9 for every positive odd integer $n$.

### Solution 2.3

**Base case ($n = 1$).** $7^1 + 2^1 = 9$, divisible by 9.
**Inductive step.** Suppose $n = 2k + 1$ is odd and $7^{2k+1} + 2^{2k+1}$ is divisible by 9. Consider $n + 2 = 2(k + 1) + 1$, the next odd integer:

$$\begin{aligned} 7^{2k+3} + 2^{2k+3} &= 7^2 \cdot 7^{2k+1} + 2^2 \cdot 2^{2k+1} \\ &= 49 \cdot 7^{2k+1} + 4 \cdot 2^{2k+1} \\ &= 45 \cdot 7^{2k+1} + 4\big(7^{2k+1} + 2^{2k+1}\big). \end{aligned}$$

Both terms are multiples of 9: the first is obvious, and the second is 4 times the inductive hypothesis (which is already a multiple of 9). Therefore $7^{2k+3} + 2^{2k+3}$ is divisible by 9, completing the induction on odd integers.

### Takeaways 2.3

Working modulo 9 turns the inductive step into a one-line parity check because $7 \equiv -2 \pmod 9$.

### Problem 2.4: Triangular Numbers

Given $T_1 = 1$ and $T_n = T_{n-1} + n$ for $n \geq 2$, prove that $T_n = \frac{n(n+1)}{2}$ for every $n \geq 1$.

### Solution 2.4

**Base case ($n = 1$).** $T_1 = 1$ and $\frac{1(1+1)}{2} = 1$, so the formula holds.
**Inductive step.** Assume $T_k = \frac{k(k+1)}{2}$ for some $k \geq 1$. Then

$$T_{k+1} = T_k + (k + 1) = \frac{k(k + 1)}{2} + (k + 1) = \frac{k(k + 1) + 2(k + 1)}{2} = \frac{(k + 1)(k + 2)}{2}.$$

Therefore the closed form holds for $k + 1$, completing the induction.

### Takeaways 2.4

Most recurrence relations in Extension 2 can be guessed and then confirmed with induction if you carefully add the new term in the inductive step.

### Problem 2.5: Polygon Interior Angle Sum

Prove that the sum of the interior angles of an $n$-gon is $(n - 2) \times 180°$ for all integers $n \geq 3$.

 

## Solution 2.5

**Base case** ($n = 3$)**.** A triangle has interior angles summing to $180°$, which equals $(3 - 2) \cdot 180°$.
**Inductive step.** Assume every $k$-gon ($k \geq 3$) has interior angle sum $(k - 2) \cdot 180°$. Take a $(k + 1)$-gon. Draw a diagonal from one vertex to split it into a triangle and a $k$-gon. The $k$-gon contributes $(k-2) \cdot 180°$ and the triangle contributes $180°$, totaling $(k - 2) \cdot 180° + 180° = (k - 1) \cdot 180°$, as required.

## Takeaways 2.5

Breaking polygons into a $k$-gon plus a triangle is a classic structural induction idea: reduce a statement for $k + 1$ objects to the known case for $k$.

## Problem 2.6: Sum of consecutive odd numbers

(a) Use mathematical induction to prove that for all integers $n \geq 1$,

$$\sum_{r=1}^{n} (2r - 1) = n^2.$$

(b) Hence find the value of

$$51 + 53 + 55 + \cdots + 99.$$

**Hint:** For (a) check the base case $n = 1$ and assume the formula for $n = k$, then add the $(k + 1)$st odd number $2(k + 1) - 1$. For (b) express the requested sum as a difference of two partial sums of the first $N$ odd numbers.

## Solution 2.6

For (a): base $n = 1$ gives $1 = 1^2$. Assume $\sum_{r=1}^{k}(2r - 1) = k^2$. Then

$$\sum_{r=1}^{k+1} (2r - 1) = k^2 + 2(k + 1) - 1 = k^2 + 2k + 1 = (k + 1)^2.$$

For (b): note $51 + \cdots + 99 = \sum_{r=1}^{50}(2r - 1) - \sum_{r=1}^{25}(2r - 1) = 50^2 - 25^2 = 2500 - 625 = 1875$.

## Takeaways 2.6

- The sum of the first $n$ odd numbers equals $n^2$; visual proofs via gnomons illustrate this nicely.
- Many evaluation problems reduce to re-indexing or subtracting partial sums.

 

## 2.2 Medium Induction Problems

**Problem 2.7: Bounding a Basel-type Sum**

Given that for every $k \geq 1$,

$$\frac{1}{(k+1)^2} - \frac{1}{k} + \frac{1}{k+1} < 0,$$

prove by induction that for all integers $n \geq 2$,

$$\frac{1}{1^2} + \frac{1}{2^2} + \cdots + \frac{1}{n^2} < 2 - \frac{1}{n}.$$

**Solution 2.7**

Define $S_n = \sum_{k=1}^{n} \frac{1}{k^2}$. The given inequality can be rearranged into

$$\frac{1}{k^2} - \left(\frac{1}{k} - \frac{1}{k+1}\right) < \frac{1}{k^2} - \left(\frac{1}{k} - \frac{1}{k+1}\right) + \frac{1}{(k+1)^2} = \frac{1}{(k+1)^2},$$

which justifies the inductive step below.

**Base case ($n = 2$).** $S_2 = 1 + \frac{1}{4} = \frac{5}{4} < \frac{3}{2} = 2 - \frac{1}{2}$, so the statement holds.

**Inductive step.** Assume $S_n < 2 - \frac{1}{n}$ for some $n \geq 2$. Applying the given inequality with $k = n$ gives $\frac{1}{(n+1)^2} < \frac{1}{n} - \frac{1}{n+1}$. Hence

$$S_{n+1} = S_n + \frac{1}{(n+1)^2} < \left(2 - \frac{1}{n}\right) + \left(\frac{1}{n} - \frac{1}{n+1}\right) = 2 - \frac{1}{n+1}.$$

Thus the inequality holds for $n+1$, completing the induction.

**Takeaways 2.7**

Induction inequalities often require massaging the inductive hypothesis into an expression that absorbs the new term; algebraic manipulation of rational expressions keeps reasoning transparent for Extension 2 students.

**Problem 2.8: Comparing $\sqrt{n!}$ and $2^n$**

Show that $\sqrt{n!} > 2^n$ for all integers $n \geq 9$.

**Solution 2.8**

It is easier to square both sides and prove $n! > 4^n$.

**Base case ($n = 9$).** $9! = 362880$ and $4^9 = 262144$, so the inequality holds.

**Inductive step.** Assume $k! > 4^k$ for some $k \geq 9$. Then

$$(k+1)! = (k+1) \cdot k! > (k+1) \cdot 4^k.$$

Because $k + 1 \geq 10$, we have $(k+1) \cdot 4^k \geq 10 \cdot 4^k = 4 \cdot (2.5) \cdot 4^k > 4 \cdot 4^k = 4^{k+1}$. Thus $(k+1)! > 4^{k+1}$, completing the induction. Taking square roots recovers the stated inequality $\sqrt{n!} > 2^n$.

 

**Takeaways 2.8**

When factorials are compared with exponentials, work with squared (or unsquared) versions that eliminate radicals and keep all quantities positive.

**Problem 2.9: Solving a Linear Recurrence**

Suppose $T_1 = 3$ and $T_n = 2T_{n-1} + 2 - n$ for $n \geq 2$. Prove that $T_n = 2^n + n$ for all $n \geq 1$.

**Solution 2.9**

**Base case ($n = 1$).** $T_1 = 3 = 2^1 + 1$, so the formula works.
**Inductive step.** Assume $T_k = 2^k + k$ for some $k \geq 1$. Then

$$\begin{aligned} T_{k+1} &= 2T_k + 2 - (k+1) \\ &= 2(2^k + k) + 1 - k \\ &= 2^{k+1} + k + 1 \\ &= 2^{k+1} + (k+1), \end{aligned}$$

which agrees with the claimed closed form. Thus the identity holds for all positive integers $n$.

**Takeaways 2.9**

Linear recurrences with constant coefficients are tailor-made for induction proofs once you conjecture the closed form by spotting patterns in the first few terms.

**Problem 2.10: Derivative of $x^n$**

Using the product rule and induction, prove that for every integer $n \geq 1$,

$$\frac{d}{dx}(x^n) = nx^{n-1}.$$

**Solution 2.10**

**Base case ($n = 1$).** $\frac{d}{dx}(x) = 1$, which equals $1 \cdot x^0$.
**Inductive step.** Assume $\frac{d}{dx}(x^k) = kx^{k-1}$. Then for $n = k + 1$,

$$x^{k+1} = x \cdot x^k.$$

By the product rule,

$$\begin{aligned} \frac{d}{dx}(x^{k+1}) &= \frac{d}{dx}(x) \cdot x^k + x \cdot \frac{d}{dx}(x^k) = 1 \cdot x^k + x \cdot kx^{k-1} \\ &= x^k + kx^k = (k+1)x^k, \end{aligned}$$

which equals $(k+1)x^{(k+1)-1}$. This completes the induction.

**Takeaways 2.10**

Induction can justify formulas students memorize in calculus by combining algebraic identities with differentiation rules they already know.

 

## Problem 2.11: Factorising $x^{3^n} - 1$

Use mathematical induction to prove that, for $n \geq 1$,

$$x^{3^n} - 1 = (x-1)(x^2+x+1)(x^6+x^3+1)\cdots\left(x^{2\cdot 3^{n-1}} + x^{3^{n-1}} + 1\right).$$

## Solution 2.11

**Base case ($n = 1$).** The right side becomes $(x-1)(x^2+x+1)$, which equals $x^3 - 1$ by the difference of cubes identity.

**Inductive step.** Assume the factorisation holds for $n = k$:

$$x^{3^k} - 1 = (x-1)(x^2+x+1)\cdots\left(x^{2\cdot 3^{k-1}} + x^{3^{k-1}} + 1\right).$$

For $n = k+1$,

$$x^{3^{k+1}} - 1 = (x^{3^k})^3 - 1 = (x^{3^k} - 1)\left(x^{2\cdot 3^k} + x^{3^k} + 1\right).$$

Replace the first factor by the induction hypothesis to obtain the same product as before with one additional term $x^{2\cdot 3^k} + x^{3^k} + 1$, giving the desired factorisation.

## Takeaways 2.11

Recognising "difference of cubes" inside the inductive step is a powerful pattern for algebraic factorizations involving rapidly growing exponents.

## Problem 2.12: Weighted geometric sum

Prove that for every integer $n \geq 1$,

$$\sum_{k=1}^{n} k2^k = (n-1)2^{n+1} + 2.$$

**Hint:**

- Verify the base case $n = 1$ first.
- For induction, add the term $(m+1)2^{m+1}$ to the assumed sum for $m$.
- (Alternative) Differentiate the finite geometric series $\sum_{k=0}^{n} x^k$ with respect to $x$ and substitute $x = 2$.

 

## Solution 2.12

**Induction.** For $n = 1$ the left-hand side is $1 \cdot 2 = 2$ and the right-hand side is $(1-1)2^2 + 2 = 2$, so the base case holds.
Assume the identity holds for $n = m \geq 1$. Then

$$\begin{aligned}\sum_{k=1}^{m+1} k2^k &= \left(\sum_{k=1}^{m} k2^k\right) + (m+1)2^{m+1} \\ &= (m-1)2^{m+1} + 2 + (m+1)2^{m+1} \\ &= m2^{m+2} + 2 = \big((m+1)-1\big)2^{(m+1)+1} + 2,\end{aligned}$$

which is the desired formula for $n = m+1$. By induction the formula holds for all $n \geq 1$.
**Proof with Calculus (Differentiation)**
Consider the geometric series:

$$P(x) = \sum_{k=0}^{n} x^k = \frac{x^{n+1}-1}{x-1}$$

Differentiating both sides with respect to $x$:

$$\frac{d}{dx}\sum_{k=0}^{n} x^k = \sum_{k=1}^{n} kx^{k-1}$$

Using the quotient rule on the right-hand side:

$$\sum_{k=1}^{n} kx^{k-1} = \frac{(n+1)x^n(x-1) - (x^{n+1}-1)}{(x-1)^2}$$

Multiply both sides by $x$ to obtain the desired summation form:

$$\sum_{k=1}^{n} kx^{k} = \frac{(n+1)x^{n+1}(x-1) - x(x^{n+1}-1)}{(x-1)^2}$$

Substitute $x = 2$:

$$\begin{aligned}\sum_{k=1}^{n} k2^k &= \frac{(n+1)2^{n+1}(2-1) - 2(2^{n+1}-1)}{(2-1)^2} \\ &= (n+1)2^{n+1} - 2\cdot 2^{n+1} + 2 \\ &= 2^{n+1}(n+1-2) + 2 \\ &= (n-1)2^{n+1} + 2\end{aligned}$$

□

## Takeaways 2.12

- Many weighted sums $\sum ka^k$ can be handled by induction or by differentiating a geometric series.
- Verifying base cases and clearly aligning the algebra in the inductive step keeps the argument clean for students.

## Problem 2.13: Product of odd factorials vs powers of factorials

Prove that for every integer $n \geq 1$,

$$1! \cdot 3! \cdot 5! \cdots (2n-1)! \geq (n!)^n.$$

 

**Hint:**

1. Write the product for $n = k+1$ as the product for $n = k$ times $(2k+1)!$ and compare with $((k+1)!)^{k+1}$.
2. Expand $(2k+1)!$ as $(2k+1)(2k)\cdots(k+2)\cdot(k+1)!$ and compare termwise.

## Solution 2.13

Base case $n = 1$ holds since $1! \geq (1!)^1$. Assume true for $n = k$. Then

$$\prod_{r=1}^{k+1}(2r-1)! = \left(\prod_{r=1}^{k}(2r-1)!\right)(2k+1)! \geq (k!)^k(2k+1)!.$$

Now $(2k+1)! = (2k+1)(2k)\cdots(k+2)\cdot(k+1)!$ contains $k$ factors each at least $k+1$, so

$$(2k+1)! \geq (k+1)^k(k+1)! = (k+1)^{k+1}k!.$$

Combining gives

$$\prod_{r=1}^{k+1}(2r-1)! \geq (k!)^k \cdot (k+1)^{k+1}k! = ((k+1)!)^{k+1},$$

completing the induction.

## Takeaways 2.13

- For product-type induction, compare multiplicative growth by isolating the new factor and estimating it against the required power.
- Counting how many terms exceed a threshold (here $k+1$) is an effective lower-bound technique.

## Problem 2.14: Bounding sums of cubes

Prove that for every integer $n \geq 2$,

$$\sum_{r=1}^{n-1} r^3 < \frac{n^4}{4} < \sum_{r=1}^{n} r^3.$$

**Hint:**

1. Check the base case $n = 2$ explicitly.
2. For the inductive step compare the difference between $\frac{(k+1)^4}{4}$ and $\frac{k^4}{4}$ with $k^3$ and $(k+1)^3$ as needed.
3. Use the binomial expansion of $(k+1)^4$ to rearrange terms.

 

## Solution 2.14

Base case $n = 2$: $1 = \sum_{r=1}^{1} r^3 < 4 = \frac{2^4}{4} < 1 + 8 = 9$. Assume the double inequality holds for $n = k \geq 2$. Then

$$\frac{(k+1)^4}{4} = \frac{k^4}{4} + k^3 + \frac{6k^2 + 4k + 1}{4}.$$

Since the extra term $\frac{6k^2+4k+1}{4} > 0$, we have $\frac{(k+1)^4}{4} > \frac{k^4}{4} + k^3$, and by the inductive hypothesis $\frac{k^4}{4} + k^3 > \sum_{r=1}^{k} r^3$. Thus the left inequality for $k+1$ holds. For the right inequality, compare $\frac{(k+1)^4}{4}$ with $\frac{k^4}{4} + (k+1)^3$ and observe the difference equals $\frac{6k^2+8k+3}{4} > 0$, giving the required result.

## Takeaways 2.14

- Verify base cases carefully when working with strict inequalities.
- **Bounding Sums:** This technique proves that a sum of cubes is bounded by the integral of $x^3$.
- **Algebraic Dominance:** In inequality induction, the goal is often to show that the "added part" of the function grows faster or slower than the "added part" of the sum.

## Problem 2.15: Product of odd factorials inequality

Prove that for every integer $n \geq 1$,

$$1! \cdot 3! \cdot 5! \cdots (2n-1)! \geq (n!)^n.$$

**Hint:**

- Compare the product for $n = k+1$ with the product for $n = k$ and isolate the new factor $(2k+1)!$.
- Expand $(2k+1)!$ as $(2k+1)(2k)\cdots(k+2)(k+1)!$ and lower-bound the first $k$ factors by $(k+1)$.

## Solution 2.15

Base case $n = 1$ is trivial. Assume the inequality for $n = k$. Then

$$\prod_{r=1}^{k+1}(2r-1)! = \left(\prod_{r=1}^{k}(2r-1)!\right)(2k+1)! \geq (k!)^k(2k+1)!.$$

Since $(2k+1)! = (2k+1)(2k)\cdots(k+2)\cdot(k+1)!$ and each of the $k$ factors in $(2k+1)\ldots(k+2)$ is at least $k+1$, we have

$$(2k+1)! \geq (k+1)^k(k+1)! = ((k+1)!)^{k+1}/(k!)^k.$$

Combining yields the claim for $k+1$, completing the induction.

 

### Takeaways 2.15

- Product-type induction frequently uses termwise bounds and counting arguments to compare multiplicative growth.
- Carefully isolate the new multiplicative factor when moving from $k$ to $k+1$.

### Problem 2.16: Bounding cubic sums

Let $a_1, a_2, \ldots, a_n$ be positive real numbers with $a_1 + a_2 + \cdots + a_n = 1$. Prove that

$$\sum_{i=1}^{n} a_i^3 \geq \frac{1}{n^2}.$$

**Hint:** Use Holder's inequality or the power mean inequality; compare $(\sum a_i)^3$ with $n^2 \sum a_i^3$.

### Solution 2.16

By Hölder (or power mean),

$$(\sum_{i=1}^{n} a_i)^3 \leq n^2 \sum_{i=1}^{n} a_i^3.$$

Since the left side equals 1, rearrange to get $\sum a_i^3 \geq 1/n^2$, as desired.

### Takeaways 2.16

- Hölder and power mean inequalities give quick bounds relating different power sums.
- Normalizing sums to 1 simplifies many symmetric inequalities.

### Problem 2.17: Reduction of the Tangent Integral

Let $I_n = \displaystyle\int_0^{\pi/4} \tan^n x \, dx$ for $n = 0, 1, 2, \ldots$.

1. Show that for $n \geq 2$,
$$I_n + I_{n-2} = \frac{1}{n-1}.$$
2. Hence find the exact value of $I_4$.

**Hint:** Write $\tan^n x = \tan^{n-2} x \cdot \tan^2 x$ and use the identity $\tan^2 x = \sec^2 x - 1$; substitute $u = \tan x$ for the term with $\sec^2 x$.

 

## Solution 2.17

**Direct Proof**
For $n \geq 2$,

$$I_n = \int_0^{\pi/4} \tan^{n-2} x(\sec^2 x - 1)\, dx = \left[\frac{\tan^{n-1} x}{n-1}\right]_0^{\pi/4} - I_{n-2} = \frac{1}{n-1} - I_{n-2}.$$

Rearranging gives $I_n + I_{n-2} = 1/(n-1)$. To find $I_4$, note

$$I_4 + I_2 = \tfrac{1}{3},$$
$$I_2 + I_0 = 1, \quad I_0 = \int_0^{\pi/4} 1\, dx = \tfrac{\pi}{4}.$$

Thus $I_2 = 1 - \frac{\pi}{4}$ and $I_4 = \frac{1}{3} - I_2 = \frac{\pi}{4} - \frac{2}{3}$.

**Inductive proof of the recurrence**
Two cases arise depending on the parity of $n$.
First, let's prove the case for even $n = 2m$. Similarly to above, we can verify the base case of $m = 1$.
Now assume for some $m \geq 1$ that for $n = 2m$ the identity $I_{2m} + I_{2m-2} = 1/(2m-1)$ holds.
Consider $n = 2m + 2$:

$$\begin{aligned} I_{2m+2} &= \int_0^{\pi/4} \tan^{2m}(x) \tan^2 x\, dx \\ &= \int_0^{\pi/4} \tan^{2m} x(\sec^2 x - 1)\, dx \\ &= \left[\frac{\tan^{2m+1} x}{2m+1}\right]_0^{\pi/4} - I_{2m} \\ &= \frac{1}{2m+1} - I_{2m}. \end{aligned}$$

Rearranging gives $I_{2m+2} + I_{2m} = 1/(2m+1)$, completing the induction step. An analogous parity-shifted argument covers odd indices, so the recurrence holds for all $n \geq 2$. A side note: The direct proof is more straightforward in this case, but the inductive approach illustrates how to handle reduction formulae recursively, though not necessry here.

## Takeaways 2.17

- Reduction formulae convert higher-power integrals into lower-power ones, enabling recursive evaluation.
- Substitution after using trigonometric identities is a common pattern.

## Problem 2.18: Logarithmic Reduction and Closed Form

Let $I_n = \displaystyle\int_1^e (\ln x)^n\, dx$ for $n \geq 0$.

1. Use integration by parts to show $I_n = e - nI_{n-1}$ for $n \geq 1$.
2. Prove by induction that

$$I_n = (-1)^n n! + e \sum_{r=0}^{n} (-1)^{n-r} \frac{n!}{r!} \quad \text{for } n \geq 0.$$

**Hint:** For part (1) take $u = (\ln x)^n$ and $dv = dx$. For part (2) verify base $I_0$ and substitute the recurrence into the inductive step.

 

**Solution 2.18**

Integration by parts with $u = (\ln x)^n$, $dv = dx$ gives $I_n = [x(\ln x)^n]_1^e - n\int_1^e (\ln x)^{n-1}\,dx = e - nI_{n-1}$. The closed form follows by induction using this recurrence and the base value $I_0 = \int_1^e 1\,dx = e$.

**Takeaways 2.18**

- Reduction formulae from integration by parts often yield closed forms when iterated.
- Combining recurrences with factorial manipulations is a useful technique for exact evaluations.

## 2.3 Advanced Induction Problems

**Problem 2.19: Recursive sequence via $f(x) = 2^x + 2^{-x}$**

Let the function $f(x)$ be defined by

$$f(x) = 2^x + 2^{-x}$$

for all real $x$.
Consider the sequence $(u_n)$ defined by

$$u_1 = \tfrac{5}{2}, \qquad u_{n+1} = \sqrt{2 + u_n} \quad (n \geq 1).$$

(i) Show that $f(x)^2 = f(2x) + 2$, and deduce that

$$f\left(\frac{x}{2}\right) = \sqrt{2 + f(x)} \quad (\forall x \in \mathbb{R}).$$

(ii) Use induction to prove that for every integer $n \geq 1$,

$$u_n = f\left(\frac{1}{2^{n-1}}\right).$$

(iii) Find the exact value of

$$\lim_{n\to\infty} 2^n\sqrt{u_n - 2}.$$

(iv) Evaluate the telescoping product limit

$$\lim_{n\to\infty} \frac{u_1u_2\cdots u_n}{2^n}.$$

**Hint:**

- Expand $(2^x + 2^{-x})^2$ and take the positive square root (note $f > 0$).
- For induction, set $x = 2^{1-k}$ when passing from $k$ to $k+1$ and use part (i).
- Write $u_n - 2 = (2^t - 2^{-t})^2$ for a small $t$ and use the limit $\lim_{t\to 0}\frac{2^t-1}{t} = \ln 2$.
- Multiply the product by the conjugate factors $2^x - 2^{-x}$ to telescope.

 

## Solution 2.19

Part (i): Expand to get $[f(x)]^2 = 2^{2x} + 2^{-2x} + 2 = f(2x) + 2$. Replace $x$ by $x/2$ and take the positive root to obtain the identity claimed.
Part (ii): Base case $u_1 = f(1) = 2 + 2^{-1} = 5/2$. If $u_k = f(2^{1-k})$ then

$$u_{k+1} = \sqrt{2 + u_k} = \sqrt{2 + f(2^{1-k})} = f\left(\frac{2^{1-k}}{2}\right) = f(2^{-k}),$$

so the statement holds by induction.
Part (iii): Put $h_n = 1/2^{n-1}$ so $u_n = 2^{h_n} + 2^{-h_n}$. Then

$$\sqrt{u_n - 2} = 2^{h_n/2} - 2^{-h_n/2}, \quad \text{where } h_n/2 = 1/2^n.$$

Thus

$$2^n\sqrt{u_n - 2} = \frac{2^t - 2^{-t}}{t}\Big|_{t=1/2^n} \to (\ln 2) - (-\ln 2) = 2\ln 2.$$

Part (iv): Let $x_k = 1/2^{k-1}$ and set $g_k = 2^{x_k} - 2^{-x_k}$. Note

$$u_k g_k = 2^{2x_k} - 2^{-2x_k} = g_{k-1}.$$

Multiplying for $k = 1, \ldots, n$ gives $P_n g_n = g_0$ where $P_n = \prod_{k=1}^n u_k$ and $g_0 = 2^2 - 2^{-2} = 15/4$. Hence

$$\frac{P_n}{2^n} = \frac{15/4}{2^n(2^{x_n} - 2^{-x_n})}.$$

With $t = x_n = 1/2^{n-1}$ we have $2^n(2^t - 2^{-t}) \to 2 \cdot 2\ln 2 = 4\ln 2$, so the limit equals $\dfrac{15}{16\ln 2}$.

## Takeaways 2.19

- What happens when $u_1 < 2$? Can we use trigonometric functions instead of exponentials?
- When $u_1 = 2$, the sequence is constant: $u_n = 2$ for all $n$ and we call the point a fixed point.
- Link this problem to the hyperbolic cosine function $\cosh x = \frac{e^x + e^{-x}}{2}$.
- Recognise when sequences stem from a two-term exponential identity such as $2^x + 2^{-x}$ (a hyperbolic/cosh-like form).
- Telescoping products often require multiplying by conjugates; identify the right companion factor.
- Convert limits with $n \to \infty$ and exponentials to standard derivative-form limits via substitution $t \to 0$.
- Using L'Hopital's rule can help evaluate tricky limits.

  L'Hopital's rule states that if $\lim_{x\to a} f(x) = 0$ and $\lim_{x\to a} g(x) = 0$, and the derivatives $f'(x)$ and $g'(x)$ are continuous near $a$ with $g'(x) \neq 0$ for $x \neq a$, then

  $$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$$

  provided the limit on the right exists.

 

## Problem 2.20: De Moivre's Theorem

Let $z = r(\cos\theta + i\sin\theta)$ with $r > 0$. Prove by induction that for every integer $n \geq 1$,

$$z^n = r^n(\cos n\theta + i\sin n\theta).$$

## Solution 2.20

**Base case ($n = 1$).** Trivial because $z^1 = r(\cos\theta + i\sin\theta)$.
**Inductive step.** Assume $z^k = r^k(\cos k\theta + i\sin k\theta)$. Then

$$\begin{aligned} z^{k+1} &= z \cdot z^k = r(\cos\theta + i\sin\theta) \cdot r^k(\cos k\theta + i\sin k\theta) \\ &= r^{k+1}\big[(\cos\theta\cos k\theta - \sin\theta\sin k\theta) + i(\sin\theta\cos k\theta + \cos\theta\sin k\theta)\big] \\ &= r^{k+1}\big[\cos((k+1)\theta) + i\sin((k+1)\theta)\big], \end{aligned}$$

using the addition formulas for sine and cosine. The statement follows by induction.

## Takeaways 2.20

Complex-number induction proofs nearly always hinge on angle addition formulas once you factor out magnitudes.

## Problem 2.21: Tiling a Defective $2^n \times 2^n$ Board

Prove that any $2^n \times 2^n$ chessboard with one square missing can be completely tiled with L-shaped trominoes for all integers $n \geq 1$.

## Solution 2.21

**Base case ($n = 1$).** A $2 \times 2$ board with one square removed exactly matches a single L-shaped tromino.
**Inductive step.** Assume the claim is true for $n = k$: every $2^k \times 2^k$ board with one missing square can be tiled. Consider a board of size $2^{k+1} \times 2^{k+1}$. Divide it into four quadrants, each of size $2^k \times 2^k$. One quadrant already has the missing square. Place a single L-shaped tromino at the center to cover one square from each of the other three quadrants, effectively creating an artificial "missing" square inside each. Now each quadrant is a $2^k \times 2^k$ board with one square missing, so by the inductive hypothesis all four quadrants can be tiled. Thus the entire board can be tiled, completing the induction.

## Takeaways 2.21

Structural induction typically requires creating the same "defect" in each subproblem so the hypothesis applies uniformly.

 

## Problem 2.22: Evaluating $I_n$ from a Recurrence

Given $I_0 = 1$ and the recurrence

$$I_n = \frac{2n}{2n+1} I_{n-1} \quad (n \geq 1),$$

show that

$$I_n = \frac{2^{2n}(n!)^2}{(2n+1)!}$$

for every $n \geq 0$.

## Solution 2.22

**Base case** $(n = 0)$**.** The closed form gives $I_0 = \frac{2^0(0!)^2}{1!} = 1$, matching the definition.
**Inductive step.** Assume the result holds for $n = k$, so $I_k = \frac{2^{2k}(k!)^2}{(2k+1)!}$. Then

$$\begin{aligned} I_{k+1} &= \frac{2(k+1)}{2(k+1)+1} I_k = \frac{2(k+1)}{2k+3} \cdot \frac{2^{2k}(k!)^2}{(2k+1)!} \\ &= \frac{2^{2k+1}(k+1)(k!)^2}{(2k+3)(2k+1)!} = \frac{2^{2k+2}((k+1)!)^2}{(2k+3)!}. \end{aligned}$$

The last step follows because $(k+1)(k!)^2 = (k+1)! \cdot k!$ and $(2k+3)! = (2k+3)(2k+2)(2k+1)!$. Therefore the closed form holds for $k+1$, and by induction it is valid for all $n$.

## Takeaways 2.22

Products such as $\frac{2n}{2n+1}$ often telescope into factorials; writing several terms explicitly helps you guess the factorial pattern to confirm by induction.

## Problem 2.23: Integer Coefficients in $\int_0^1 x^n e^x \, dx$

Let

$$I_n = \int_0^1 x^n e^x \, dx$$

.
Show by induction that there exist integers $a_n$ and $b_n$ with $I_n = a_n + b_n e$ for every $n \geq 0$.

## Solution 2.23

Integrate by parts with $u = x^n$ and $dv = e^x dx$ to obtain, for $n \geq 1$,

$$I_n = [x^n e^x]_0^1 - n \int_0^1 x^{n-1} e^x dx = e - nI_{n-1}.$$

**Base case** $(n = 0)$**.** $I_0 = \int_0^1 e^x dx = e - 1$, so choose $(a_0, b_0) = (-1, 1)$.
**Inductive step.** Suppose $I_{n-1} = a_{n-1} + b_{n-1}e$ with integers $a_{n-1}, b_{n-1}$. Then

$$I_n = e - n(a_{n-1} + b_{n-1}e) = (-na_{n-1}) + (1 - nb_{n-1})e.$$

Both coefficients are integers, so set $a_n = -na_{n-1}$ and $b_n = 1 - nb_{n-1}$. Therefore $I_n$ always takes the form $a_n + b_n e$ with integers $a_n, b_n$.

 

**Takeaways 2.23**

The recurrence $I_n = e - nI_{n-1}$ keeps the structure "integer plus integer times $e$," so tracking the coefficients directly is an efficient inductive strategy.

**Problem 2.24: Closed Form for $J_n$**

Let $J_n = \displaystyle\int_0^1 x^n e^{-x}\,dx$ with $J_0 = 1 - \frac{1}{e}$. Show that for every $n \geq 0$,

$$J_n = n! - \frac{n!}{e}\sum_{r=0}^{n}\frac{1}{r!}.$$

**Solution 2.24**

We have the recurrence $J_n = nJ_{n-1} - \frac{1}{e}$ for $n \geq 1$.
**Base case** ($n = 0$)**.** The right-hand side yields $0! - \frac{0!}{e} \cdot 1 = 1 - \frac{1}{e}$, which equals $J_0$.
**Inductive step.** Assume $J_k = k! - \frac{k!}{e}\sum_{r=0}^{k}\frac{1}{r!}$. Then

$$\begin{aligned}
J_{k+1} &= (k+1)J_k - \frac{1}{e} \\
&= (k+1)\left(k! - \frac{k!}{e}\sum_{r=0}^{k}\frac{1}{r!}\right) - \frac{1}{e} \\
&= (k+1)! - \frac{(k+1)!}{e}\sum_{r=0}^{k}\frac{1}{r!} - \frac{1}{e}.
\end{aligned}$$

Observe that $\frac{1}{e} = \frac{(k+1)!}{e} \cdot \frac{1}{(k+1)!}$. Therefore

$$J_{k+1} = (k+1)! - \frac{(k+1)!}{e}\left(\sum_{r=0}^{k}\frac{1}{r!} + \frac{1}{(k+1)!}\right) = (k+1)! - \frac{(k+1)!}{e}\sum_{r=0}^{k+1}\frac{1}{r!}.$$

This matches the claimed closed form, completing the induction.

**Takeaways 2.24**

Recurrences derived from integration by parts usually reveal factorial patterns; substituting the hypothesis and carefully aligning summations keep the algebra manageable.

# 3 Part 2: Problems and Solutions (Concise + Hints)

Part 2 revisits three new sets of problems. Solutions are intentionally briefer to encourage student ownership, and every problem includes an upside-down hint.

## 3.1 Basic Induction Problems

**Problem 3.1: Sum of First $n$ Odd Numbers**

Prove that $1 + 3 + 5 + \cdots + (2n-1) = n^2$ for all integers $n \geq 1$.

 

**Hint:** The $(n+1)$-st odd number is $2(n+1)-1=2n+1$. Add this to the hypothesis $1+3+\cdots+(2n-1)=n^2$ and factor the result.

### Solution 3.1: Sketch

Base case: $n=1$ gives $1=1^2$. For the inductive step, assume the sum of the first $k$ odd numbers equals $k^2$. Then the sum of the first $k+1$ odd numbers is $k^2+(2k+1)=k^2+2k+1=(k+1)^2$.

### Problem 3.2: Divisibility of $4^n-1$ by 3

Prove that $4^n-1$ is divisible by 3 for every integer $n \geq 1$.

**Hint:** Note that $4 \equiv 1 \pmod 3$, so $4^{k+1}-1=4\cdot 4^k-1=4(4^k-1)+3$.

### Solution 3.2: Sketch

Base case: $4^1-1=3$ is divisible by 3. If $4^k-1$ is divisible by 3, then $4^{k+1}-1=4\cdot 4^k-1=4(4^k-1)+3$, which is the sum of two multiples of 3.

### Problem 3.3: Sum of First $n$ Squares

Prove that

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$$

for all integers $n \geq 1$.

**Hint:** Add $(k+1)^2$ to both sides of the hypothesis and show the result simplifies to $\frac{(k+1)(k+2)(2k+3)}{6}$.

### Solution 3.3: Sketch

Base case: $1^2=1=\frac{1\cdot 2\cdot 3}{6}$. Assume the formula holds for $n=k$. Then $\sum_{r=1}^{k+1} r^2 = \frac{k(k+1)(2k+1)}{6}+(k+1)^2 = \frac{k(k+1)(2k+1)+6(k+1)^2}{6} = \frac{(k+1)(2k^2+7k+6)}{6} = \frac{(k+1)(k+2)(2k+3)}{6}$.

### Problem 3.4: Geometric Series

Prove that

$$\sum_{k=0}^{n} r^k = \frac{r^{n+1}-1}{r-1}$$

for all integers $n \geq 0$ and $r \neq 1$.

 

**Hint:** Multiply both sides of the hypothesis by $r$ and compare with the $(n+1)$ case to derive a telescoping relationship.

**Solution 3.4: Sketch**

Base case: $r^0 = 1 = \frac{r-1}{r-1}$. Assume the formula holds for $n = k$. Then $\sum_{j=0}^{k+1} r^j = \frac{r^{k+1}-1}{r-1} + r^{k+1} = \frac{r^{k+1}-1+r^{k+1}(r-1)}{r-1} = \frac{r^{k+2}-1}{r-1}$.

**Problem 3.5: Bernoulli's Inequality**

Prove that $(1+x)^n \geq 1+nx$ for all real numbers $x > -1$ and integers $n \geq 1$.

**Hint:** Multiply the hypothesis $(1+x)^k \geq 1+kx$ by $(1+x)$ and use the fact that $x^2 \geq 0$ and $kx^2 \geq 0$ when $k \geq 1$.

**Solution 3.5: Sketch**

Base case: $(1+x)^1 = 1+x$. Assume $(1+x)^k \geq 1+kx$ for some $k \geq 1$. Then $(1+x)^{k+1} = (1+x)(1+x)^k \geq (1+x)(1+kx) = 1+kx+x+kx^2 = 1+(k+1)x+kx^2 \geq 1+(k+1)x$ since $kx^2 \geq 0$.

**Problem 3.6: De Moivre's Formula (Conjugate Form)**

Prove by induction that for all integers $n \geq 1$:

$$(\cos\theta - i\sin\theta)^n = \cos(n\theta) - i\sin(n\theta)$$

**Hint:** Use the standard angle addition formulas: $\cos(A+B) = \cos A\cos B - \sin A\sin B$ and $\sin(A+B) = \sin A\cos B + \cos A\sin B$.

**Solution 3.6: Sketch**

Base case: $n=1$ gives $\cos\theta - i\sin\theta = \cos\theta - i\sin\theta$. Assume the formula holds for $n=k$. Then $(\cos\theta - i\sin\theta)^{k+1} = (\cos k\theta - i\sin k\theta)(\cos\theta - i\sin\theta) = \cos k\theta\cos\theta - \sin k\theta\sin\theta - i(\sin k\theta\cos\theta + \cos k\theta\sin\theta) = \cos(k+1)\theta - i\sin(k+1)\theta$.

**Problem 3.7: Power Inequality $2^n \geq n^2 - 2$**

Prove by induction that $2^n \geq n^2 - 2$ for all integers $n \geq 3$.

**Hint:** Show that $2\cdot k^2 - 2 \geq (k+1)^2 - 2$ for $k \geq 3$, which simplifies to $k^2 - 2k - 1 \geq 0$.

 

### Solution 3.7: Sketch

Base case: $n = 3$ gives $2^3 = 8 \geq 7 = 3^2 - 2$. Assume $2^k \geq k^2 - 2$ for some $k \geq 3$. Then $2^{k+1} = 2 \cdot 2^k \geq 2(k^2 - 2) = 2k^2 - 4$. We need $2k^2 - 4 \geq (k+1)^2 - 2 = k^2 + 2k - 1$, which simplifies to $k^2 - 2k - 3 \geq 0$ or $(k-3)(k+1) \geq 0$. This holds for all $k \geq 3$.

### Problem 3.8: Logarithmic derivative of a polynomial

Given the polynomial

$$P_n(x) = \prod_{i=1}^{n}(x - a_i) = (x - a_1)(x - a_2)\ldots(x - a_n),$$

prove by mathematical induction that for all integers $n \geq 2$:

$$\frac{P_n'(x)}{P_n(x)} = \frac{1}{x - a_1} + \frac{1}{x - a_2} + \cdots + \frac{1}{x - a_n}.$$

**Hint:**

- Verify the base case $n = 2$ using the product rule $(uv)' = u'v + uv'$.
- Write $P_{k+1}(x) = P_k(x)(x - a_{k+1})$ and differentiate using the product rule.
- Divide the derivative $P_{k+1}'(x)$ by $P_{k+1}(x)$ and use the inductive hypothesis.

### Solution 3.8

**Base case.** For $n = 2$, $P_2(x) = (x - a_1)(x - a_2)$ and

$$P_2'(x) = (x - a_2) + (x - a_1),$$

so

$$\frac{P_2'(x)}{P_2(x)} = \frac{(x - a_1) + (x - a_2)}{(x - a_1)(x - a_2)} = \frac{1}{x - a_1} + \frac{1}{x - a_2}.$$

**Inductive step.** Assume for some $k \geq 2$ that

$$\frac{P_k'(x)}{P_k(x)} = \sum_{i=1}^{k} \frac{1}{x - a_i}.$$

Let $P_{k+1}(x) = P_k(x)(x - a_{k+1})$. Differentiating gives

$$P_{k+1}'(x) = P_k'(x)(x - a_{k+1}) + P_k(x).$$

Divide by $P_{k+1}(x) = P_k(x)(x - a_{k+1})$ to obtain

$$\frac{P_{k+1}'(x)}{P_{k+1}(x)} = \frac{P_k'(x)}{P_k(x)} + \frac{1}{x - a_{k+1}}.$$

Substituting the inductive hypothesis yields the desired identity for $k+1$. Thus the formula holds for all $n \geq 2$.

 

**Takeaways 3.1**

- This identity is the logarithmic derivative: $\frac{d}{dx}\ln P_n(x) = \frac{P_n'(x)}{P_n(x)}$, turning products into sums. No induction required!
- It generalises the product rule to $n$ factors and is useful for partial-fraction decompositions.
- The proof is a straightforward application of the product rule combined with induction.

## 3.2 Medium Induction Problems

**Problem 3.9: Sum of Cubes**

Prove that

$$\sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2$$

for all integers $n \geq 1$.

**Hint:** Note that the right side is the square of the sum of the first $n$ positive integers. Add $(k+1)^3$ to the hypothesis and show it equals $\left(\frac{(k+1)(k+2)}{2}\right)^2$.

**Solution 3.9: Sketch**

Base case: $1^3 = 1 = \left(\frac{1\cdot 2}{2}\right)^2$. Assume the formula holds for $n = k$. Then $\sum_{r=1}^{k+1} r^3 = \left(\frac{k(k+1)}{2}\right)^2 + (k+1)^3 = \frac{k^2(k+1)^2+4(k+1)^3}{4} = \frac{(k+1)^2(k^2+4k+4)}{4} = \left(\frac{(k+1)(k+2)}{2}\right)^2$.

**Problem 3.10: Tower Inequality**

Prove that $2^n > n^2$ for all integers $n \geq 5$.

**Hint:** For the inductive step, show that $2 \cdot k^2 > (k+1)^2$ when $k \geq 5$ by expanding and simplifying.

**Solution 3.10: Sketch**

Base case: $2^5 = 32 > 25 = 5^2$. Assume $2^k > k^2$ for some $k \geq 5$. Then $2^{k+1} = 2 \cdot 2^k > 2k^2$. We need $2k^2 \geq (k+1)^2 = k^2 + 2k + 1$, which simplifies to $k^2 \geq 2k + 1$ or $k^2 - 2k - 1 \geq 0$. This holds for $k \geq 5$ since $25 - 10 - 1 = 14 > 0$.

**Problem 3.11: Inequality with Factorials**

Prove that $n! > 2^n$ for all integers $n \geq 4$, where $n! = n \times (n-1) \times (n-2) \times \cdots \times 2 \times 1$.

 

**Hint:** In the inductive step, multiply the hypothesis by $(k+1)$ and show that $(k+1)\cdot 2^k > 2^{k+1}$ when $k \geq 4$.

**Solution 3.11: Sketch**

Base case: $4! = 24 > 16 = 2^4$. Assume $k! > 2^k$ for some $k \geq 4$. Then $(k+1)! = (k+1)\cdot k! > (k+1)\cdot 2^k$. Since $k+1 \geq 5 > 2$, we have $(k+1)\cdot 2^k > 2\cdot 2^k = 2^{k+1}$.

**Problem 3.12: Postage Stamp Problem**

Prove that any integer $n \geq 12$ can be expressed as $n = 4a + 5b$ for some non-negative integers $a$ and $b$.

**Hint:** Use strong induction with base cases $n = 12, 13, 14, 15$. For $n \geq 16$, express $n = (n-4)+4$ and apply the hypothesis to $n-4 \geq 12$.

**Solution 3.12: Sketch**

Base cases: $12 = 4(3)+5(0)$, $13 = 4(2)+5(1)$, $14 = 4(1)+5(2)$, $15 = 4(0)+5(3)$. For $k \geq 16$, by hypothesis $k-4 = 4a+5b$ for some $a, b \geq 0$. Then $k = (k-4)+4 = 4(a+1)+5b$.

**Problem 3.13: Inequality $3^n > n^3$**

Prove that $3^n > n^3$ for all integers $n \geq 4$.

**Hint:** Show that $3k^3 > (k+1)^3$ for $k \geq 4$ by expanding $(k+1)^3$ and verifying $3k^3 - k^3 - 3k^2 - 3k - 1 > 0$.

**Solution 3.13: Sketch**

Base case: $3^4 = 81 > 64 = 4^3$. Assume $3^k > k^3$ for some $k \geq 4$. Then $3^{k+1} = 3\cdot 3^k > 3k^3$. We need $3k^3 > (k+1)^3 = k^3 + 3k^2 + 3k + 1$, which simplifies to $2k^3 > 3k^2 + 3k + 1$. For $k = 4$: $2(64) = 128 > 48 + 12 + 1 = 61$. Since both sides grow with $k$, the inequality holds for all $k \geq 4$.

**Problem 3.14: Factorial Inequality $(2n)! \geq 2^n(n!)^2$**

Prove that $(2n)! \geq 2^n(n!)^2$ for all positive integers $n$.

**Hint:** For the inductive step, express $(2(k+1))!$ as $(2k+2)(2k+1)\cdot(2k)!$ and show that $(2k+2)(2k+1) \geq 2(k+1)^2$ by expanding and simplifying.

 

### Solution 3.14: Sketch

Base case: $n = 1$ gives $(2 \cdot 1)! = 2 \geq 2^1(1!)^2 = 2$. Assume $(2k)! \geq 2^k(k!)^2$ for some $k \geq 1$. Then $(2(k+1))! = (2k+2)! = (2k+2)(2k+1)(2k)! \geq (2k+2)(2k+1) \cdot 2^k(k!)^2$. We need $(2k+2)(2k+1) \geq 2(k+1)^2$, or $(2k+2)(2k+1) \geq 2(k^2+2k+1)$. Expanding: $4k^2+6k+2 \geq 2k^2+4k+2$, which simplifies to $2k^2+2k \geq 0$, true for all $k \geq 1$. Thus $(2(k+1))! \geq 2^{k+1}((k+1)!)^2$.

### Problem 3.15: Exponential Inequality $4^n - 1 - 7n > 0$

Use mathematical induction to prove that $4^n - 1 - 7n > 0$ for all integers $n \geq 2$.

**Hint:** For the inductive step, show that $4(4^k - 1 - 7k) < 4^{k+1} - 1 - 7(k+1)$ reduces to $3 \cdot 4^k + 21k > 7$, which is clearly true for $k \geq 2$.

### Solution 3.15: Sketch

Base case: $n = 2$ gives $4^2 - 1 - 14 = 1 > 0$. Assume $4^k - 1 - 7k > 0$ for some $k \geq 2$. Then $4^{k+1} - 1 - 7(k+1) = 4 \cdot 4^k - 1 - 7k - 7 = 4(4^k - 1 - 7k) + 3 \cdot 4^k + 21k$. Since $4^k - 1 - 7k > 0$ by hypothesis and $3 \cdot 4^k + 21k > 0$, we have $4^{k+1} - 1 - 7(k+1) > 0$.

### Problem 3.16: Weighted Geometric Sum

Prove by induction that for integers $n \geq 1$:

$$1 + 2\left(\frac{1}{2}\right) + 3\left(\frac{1}{2}\right)^2 + \cdots + n\left(\frac{1}{2}\right)^{n-1} = 4 - \frac{n+2}{2^{n-1}}$$

**Hint:** Add the $(k+1)$-th term $(k+1)\left(\frac{1}{2}\right)^k$ to the hypothesis and show it simplifies to $4 - \frac{k+3}{2^k}$.

### Solution 3.16: Sketch

Base case: $n = 1$ gives $1 = 4 - 3$. Assume the formula holds for $n = k$. Then the sum up to $k + 1$ is $4 - \frac{k+2}{2^{k-1}} + (k+1)\left(\frac{1}{2}\right)^k = 4 - \frac{2(k+2)}{2^k} + \frac{k+1}{2^k} = 4 - \frac{2k+4-k-1}{2^k} = 4 - \frac{k+3}{2^k}$.

### Problem 3.17: Factorial Series

Prove by mathematical induction that for any positive integer $n$, the following equality holds:

$$(n+1)! = 1 + \frac{1!^2}{0!} + \frac{2!^2}{1!} + \cdots + \frac{n!^2}{(n-1)!}$$

 

**Hint:**

- **Hint 1:** Before starting the induction, simplify the general term in the series. What does $\frac{k!^2}{(k-1)!}$ simplify to? Write $k!$ as $k \cdot (k-1)!$.
- **Hint 2:** For the inductive step, assume the statement holds for $n = k$. When adding the $(k+1)$-th term to both sides, look for a common factor to pull out to achieve $(k+2)!$.

**Solution 3.17: Brief**

Let $P(n)$ be the proposition.
**Base Case ($n = 1$):** LHS $= (1+1)! = 2$. RHS $= 1 + \frac{1!^2}{0!} = 1 + 1 = 2$. $P(1)$ is true.
**Inductive Step:** Assume $P(k)$ is true for some positive integer $k$:

$$(k+1)! = 1 + \frac{1!^2}{0!} + \cdots + \frac{k!^2}{(k-1)!}$$

We want to prove $P(k+1)$. Add the $(k+1)$-th term to both sides:

$$\text{RHS of } P(k+1) = (k+1)! + \frac{(k+1)!^2}{k!}$$

Note that $\frac{(k+1)!^2}{k!} = (k+1)! \cdot \frac{(k+1)!}{k!} = (k+1)!(k+1)$.
Substituting this back in:

$$\text{RHS} = (k+1)! + (k+1)!(k+1)$$

Factor out $(k+1)!$:

$$\text{RHS} = (k+1)![1 + (k+1)] = (k+1)!(k+2) = (k+2)!$$

This is exactly the LHS of $P(k+1)$. Hence, by the principle of mathematical induction, the statement is true for all integers $n \geq 1$.

**Takeaways 3.2**

- **Algebraic simplification before induction:** Simplifying the terms of a series (e.g., recognizing factorials can cancel) drastically reduces the algebraic burden during the inductive step.
- **Factoring is your friend:** In factorial induction proofs, always look to factor out the lowest common factorial rather than expanding everything.

## 3.3 Advanced Induction Problems

**Problem 3.18: Binomial Theorem**

Prove by inductionthat

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k$$

for all integers $n \geq 0$ and real numbers $a, b$.

**Hint:** Use Pascal's identity $\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$ to expand $(a+b)^{n+1} = (a+b)(a+b)^n$.

 

**Solution 3.18: Sketch**

Base case: $(a+b)^0 = 1 = \binom{0}{0}a^0b^0$. Assume the formula holds for $n$. Then $(a+b)^{n+1} = (a+b)\sum_{k=0}^{n}\binom{n}{k}a^{n-k}b^k = \sum_{k=0}^{n}\binom{n}{k}a^{n+1-k}b^k + \sum_{k=0}^{n}\binom{n}{k}a^{n-k}b^{k+1}$. Reindex and apply Pascal's identity to combine terms.

**Problem 3.19: Fermat's Little Theorem**

For any prime $p$ and integer $a$, prove that $a^p \equiv a \pmod{p}$.

**Hint:** Use induction on $a$. Show $(k+1)^p - (k+1) \equiv k^p - k \pmod{p}$ by expanding with the binomial theorem and noting that $\binom{p}{j}$ is divisible by $p$ for $1 \le j \le p-1$.

**Solution 3.19: Sketch**

Base case: $0^p \equiv 0 \pmod{p}$. Assume $k^p \equiv k \pmod{p}$. Then $(k+1)^p = k^p + \binom{p}{1}k^{p-1} + \cdots + \binom{p}{p-1}k + 1$. Since $p \mid \binom{p}{j}$ for $1 \le j \le p-1$, we have $(k+1)^p \equiv k^p + 1 \equiv k+1 \pmod{p}$ by the hypothesis.

**Problem 3.20: Symmetric Sum Inequality**

Let $n$ be a positive integer and $a_1, a_2, \ldots, a_n$ be $n$ positive real numbers. Prove by mathematical induction that:

$$(a_1 + a_2 + \cdots + a_n)\left(\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n}\right) \ge n^2$$

for all $n \ge 1$.

**Hint:** Use induction on $n$. For the inductive step, expand $\left(\sum_{i=1}^{k+1} a_i\right)\left(\sum_{i=1}^{k+1} \frac{1}{a_i}\right)$ and apply the inequality $x + \frac{1}{x} \ge 2$ for $x > 0$ to each pair $\frac{a_i}{a_{k+1}}$ and $\frac{a_{k+1}}{a_i}$.

 

## Solution 3.20: Sketch

Base case: $n = 1$. $(a_1)\left(\frac{1}{a_1}\right) = 1 \geq 1^2$.

Inductive hypothesis: Assume true for $n = k$, i.e., $\left(\sum_{i=1}^{k} a_i\right)\left(\sum_{i=1}^{k} \frac{1}{a_i}\right) \geq k^2$.

Inductive step: For $n = k+1$,

$$\begin{aligned}\left(\sum_{i=1}^{k+1} a_i\right)\left(\sum_{i=1}^{k+1} \frac{1}{a_i}\right) &= (S_k + a_{k+1})\left(R_k + \frac{1}{a_{k+1}}\right)\\ &= S_k R_k + S_k \cdot \frac{1}{a_{k+1}} + a_{k+1} \cdot R_k + 1\\ &= S_k R_k + 1 + \sum_{i=1}^{k}\left(\frac{a_i}{a_{k+1}} + \frac{a_{k+1}}{a_i}\right)\end{aligned}$$

By the hypothesis, $S_k R_k \geq k^2$. For each $i$, $\frac{a_i}{a_{k+1}} + \frac{a_{k+1}}{a_i} \geq 2$. Thus,

$$\mathrm{LHS}_{k+1} \geq k^2 + 1 + 2k = (k+1)^2$$

So the result holds for $n = k+1$.

Conclusion: By induction, the inequality holds for all $n \geq 1$.

## Problem 3.21: Tower of Hanoi

The Tower of Hanoi is a classic puzzle involving three pegs and $n$ disks of different sizes, all initially stacked in order of decreasing size on one peg. The objective is to move the entire stack to another peg, moving only one disk at a time and never placing a larger disk on top of a smaller one. Prove that the minimum number of moves required to transfer $n$ disks in the Tower of Hanoi puzzle is $2^n - 1$.

**Hint:** To move $k+1$ disks, first move the top $k$ disks to the auxiliary peg (requiring $T_k$ moves), then move the largest disk (1 move), then move the $k$ disks to the destination (requiring $T_k$ moves again). This gives $T_{k+1} = 2T_k + 1$.

## Solution 3.21: Sketch

Base case: $T_1 = 1 = 2^1 - 1$. Assume $T_k = 2^k - 1$. To move $k+1$ disks: move $k$ disks to auxiliary peg ($2^k - 1$ moves), move largest disk (1 move), move $k$ disks to destination ($2^k - 1$ moves). Total: $T_{k+1} = 2(2^k - 1) + 1 = 2^{k+1} - 1$.

## Problem 3.22: Vandermonde's Identity

Prove that $\binom{m+n}{r} = \sum_{k=0}^{r} \binom{m}{k}\binom{n}{r-k}$ for non-negative integers $m, n, r$.
Here, by definition, $\binom{a}{b} = \frac{a!}{b!(a-b)!}$ for integers $0 \leq b \leq a$, and $\binom{a}{b} = 0$ if $b > a$.

**Hint:** Use induction on $n$. Apply Pascal's identity $\binom{n+1}{j} = \binom{n}{j-1} + \binom{n}{j}$ to split the right-hand side.

 

### Solution 3.22: Sketch

Induct on $n$ with $m, r$ fixed. Base case $n = 0$: both sides equal $\binom{m}{r}$. Assume the identity holds for $n$. Then $\binom{m+n+1}{r} = \binom{m+n}{r-1} + \binom{m+n}{r}$. Apply the hypothesis to both terms and rearrange using $\binom{n}{j-1} + \binom{n}{j} = \binom{n+1}{j}$ to obtain the desired sum.

### Problem 3.23: Wilson's Theorem

Prove that $(p-1)! \equiv -1 \pmod{p}$ for every prime $p$.

**Hint:**For $p > 2$, pair each $a \in \{2, 3, \ldots, p-2\}$ with its inverse $a^{-1} \pmod{p}$. The only elements that are their own inverses are 1 and $p-1$.

### Solution 3.23: Sketch

Base case: $p = 2$ gives $(2-1)! = 1 \equiv -1 \pmod{2}$. For prime $p > 2$, in the product $(p-1)! = 1 \cdot 2 \cdots (p-1)$, each $a \in \{2, \ldots, p-2\}$ pairs with its inverse $a^{-1}$ (where $a \neq a^{-1}$), contributing $aa^{-1} \equiv 1$. Only 1 and $p-1$ are self-inverse, so $(p-1)! \equiv 1 \cdot (p-1) \equiv -1 \pmod{p}$.

### Problem 3.24: Recursive Sequence with Surds

A sequence $a_n$ is defined by $a_n = 2a_{n-1} + a_{n-2}$ for $n \geq 2$ with $a_0 = a_1 = 2$. Use mathematical induction to prove that:

$$a_n = (1+\sqrt{2})^n + (1-\sqrt{2})^n \quad \text{for all } n \geq 0$$

**Hint:**Use strong induction. For the base cases verify $n = 0$ and $n = 1$. Then show $(1+\sqrt{2})^{k+1} + (1-\sqrt{2})^{k+1} = 2[(1+\sqrt{2})^k + (1-\sqrt{2})^k] + [(1+\sqrt{2})^{k-1} + (1-\sqrt{2})^{k-1}]$ using the fact that $(1 \pm \sqrt{2})$ satisfy $x^2 = 2x + 1$.

### Solution 3.24: Sketch

Base cases: $a_0 = 2 = 1 + 1$, $a_1 = 2 = (1+\sqrt{2}) + (1-\sqrt{2})$. Assume the formula holds for $n = k-1$ and $n = k$. Note that $\alpha = 1 + \sqrt{2}$ and $\beta = 1 - \sqrt{2}$ both satisfy $x^2 = 2x + 1$. Then $a_{k+1} = 2a_k + a_{k-1} = 2(\alpha^k + \beta^k) + (\alpha^{k-1} + \beta^{k-1}) = \alpha^{k-1}(2\alpha + 1) + \beta^{k-1}(2\beta + 1) = \alpha^{k-1} \cdot \alpha^2 + \beta^{k-1} \cdot \beta^2 = \alpha^{k+1} + \beta^{k+1}$.

### Problem 3.25: Powers of $2 + \sqrt{3}$

For each positive integer $n$, prove that there exist unique positive integers $p_n$ and $q_n$ such that

$$(2+\sqrt{3})^n = p_n + q_n\sqrt{3}.$$

Then show that these numbers satisfy the identity $p_n^2 - 3q_n^2 = 1$ for all $n$.

 

**Hint:** Induct on $n$. For the step $k \to k+1$, expand $(2+\sqrt{3})^{k+1} = (2+\sqrt{3})^k(2+\sqrt{3})$, substitute the hypothesis $(2+\sqrt{3})^k = p_k + q_k\sqrt{3}$, and collect rational and irrational parts to define $p_{k+1}$ and $q_{k+1}$. For the identity, consider the conjugate $(2-\sqrt{3})^n = p_n - q_n\sqrt{3}$ and multiply the two expressions.

## Solution 3.25: Sketch

Base case $n = 1$: $(2+\sqrt{3}) = 2 + 1\cdot\sqrt{3}$, so $p_1 = 2$, $q_1 = 1$. Inductive step: assume $(2+\sqrt{3})^k = p_k + q_k\sqrt{3}$. Then

$$\begin{aligned}(2+\sqrt{3})^{k+1} &= (p_k + q_k\sqrt{3})(2+\sqrt{3})\\ &= (2p_k + 3q_k) + (p_k + 2q_k)\sqrt{3}.\end{aligned}$$

Define $p_{k+1} = 2p_k + 3q_k$ and $q_{k+1} = p_k + 2q_k$; both remain positive integers, and uniqueness follows since $\sqrt{3}$ is irrational. For the identity, note $(2-\sqrt{3})^n = p_n - q_n\sqrt{3}$ by the same recurrence. Multiplying the conjugate pair gives $[(2+\sqrt{3})(2-\sqrt{3})]^n = (p_n + q_n\sqrt{3})(p_n - q_n\sqrt{3}) = p_n^2 - 3q_n^2 = 1$.

## Takeaways 3.3

- Powers of a quadratic surd generate integer pairs via linear recurrences $p_{n+1} = 2p_n + 3q_n$, $q_{n+1} = p_n + 2q_n$.
- Conjugates neutralize the irrational part, revealing invariants like $p_n^2 - 3q_n^2 = 1$ (a Pell-type relation).
- Induction plus conjugation is a standard method for identities involving quadratic irrationals.

## Problem 3.26: Sum of Cosines

It is given that $2\cos A \sin B = \sin(A+B) - \sin(A-B)$. Prove by induction that for integers $n \geq 1$:

$$\cos\theta + \cos 3\theta + \cdots + \cos(2n-1)\theta = \frac{\sin 2n\theta}{2\sin\theta}$$

**Hint:** Use the product-to-sum identity to show that $\sin 2(k+1)\theta = \sin 2k\theta + 2\sin\theta\cos(2k+1)\theta$.

## Solution 3.26: Sketch

Base case: $n = 1$ gives $\cos\theta = \frac{\sin 2\theta}{2\sin\theta} = \frac{2\sin\theta\cos\theta}{2\sin\theta} = \cos\theta$. Assume $\sum_{r=1}^{k}\cos(2r-1)\theta = \frac{\sin 2k\theta}{2\sin\theta}$. Then $\sum_{r=1}^{k+1}\cos(2r-1)\theta = \frac{\sin 2k\theta}{2\sin\theta} + \cos(2k+1)\theta$. Using $2\cos(2k+1)\theta\sin\theta = \sin(2k+2)\theta - \sin 2k\theta$, we get $\cos(2k+1)\theta = \frac{\sin 2(k+1)\theta - \sin 2k\theta}{2\sin\theta}$. Thus the sum equals $\frac{\sin 2(k+1)\theta}{2\sin\theta}$.

 

## Problem 3.27: Nested Radicals

The numbers $a_n$, for integers $n \geq 1$, are defined as $a_1 = \sqrt{2}$, $a_2 = \sqrt{2+\sqrt{2}}$, $a_3 = \sqrt{2+\sqrt{2+\sqrt{2}}}$ and so on. These numbers satisfy the relation $a_{n+1}^2 = 2 + a_n$. Use mathematical induction to prove that:

$$a_n = 2\cos\frac{\pi}{2^{n+1}} \quad \text{for all integers } n \geq 1$$

**Hint:** Use the half-angle formula: $\cos\frac{\theta}{2} = \sqrt{\frac{1+\cos\theta}{2}}$ to show that $2\cos\frac{\pi}{2^{k+2}}$ satisfies the recurrence when squared.

## Solution 3.27: Sketch

Base case: $a_1 = \sqrt{2} = 2\cos\frac{\pi}{4}$. Assume $a_k = 2\cos\frac{\pi}{2^{k+1}}$. Then $a_{k+1}^2 = 2 + a_k = 2 + 2\cos\frac{\pi}{2^{k+1}} = 2\left(1+\cos\frac{\pi}{2^{k+1}}\right) = 4\cos^2\frac{\pi}{2^{k+2}}$ using the double-angle formula $1+\cos\theta = 2\cos^2\frac{\theta}{2}$. Since $a_{k+1} > 0$, we have $a_{k+1} = 2\cos\frac{\pi}{2^{k+2}}$.

## Problem 3.28: Cosecant Sum Formula

Use mathematical induction to prove that for all $n \geq 1$:

$$\sum_{r=1}^{n} \csc(2^r x) = \cot x - \cot(2^n x)$$

**Hint:** Use the identity $\cot x - \cot 2x = \csc 2x$ to simplify the $(k+1)$-th term.

## Solution 3.28: Sketch

Base case: $n = 1$ gives $\csc 2x = \frac{1}{\sin 2x} = \frac{1}{2\sin x\cos x} = \frac{1}{2\sin x}\cdot\frac{1}{\cos x} = \cot x - \cot 2x$ (using $\cot x - \cot 2x = \frac{\cos x}{\sin x} - \frac{\cos 2x}{\sin 2x} = \frac{2\cos x\sin x - \cos 2x\sin x}{\sin 2x\sin x} = \frac{\sin 2x}{\sin 2x\sin x} = \csc 2x$). Assume $\sum_{r=1}^{k}\csc(2^r x) = \cot x - \cot(2^k x)$. Then $\sum_{r=1}^{k+1}\csc(2^r x) = \cot x - \cot(2^k x) + \csc(2^{k+1}x) = \cot x - [\cot(2^k x) - \csc(2^{k+1}x)] = \cot x - \cot(2^{k+1}x)$.

## Problem 3.29: Arctangent Sum

Use mathematical induction to prove that for all positive integers $n$:

$$\sum_{j=1}^{n} \tan^{-1}\left(\frac{1}{2j^2}\right) = \tan^{-1}\left(\frac{n}{n+1}\right)$$

**Hint:** Use the arctangent addition formula: $\tan^{-1} a + \tan^{-1} b = \tan^{-1}\left(\frac{a+b}{1-ab}\right)$ when $ab < 1$. Show that $\frac{1}{2(k+1)^2} + \frac{k}{k+1} = \frac{k+1}{k+2}$ after applying the formula.

 

## Solution 3.29: Sketch

Base case: $n = 1$ gives $\tan^{-1}\frac{1}{2} = \tan^{-1}\frac{1}{2}$. Assume $\sum_{j=1}^{k}\tan^{-1}\left(\frac{1}{2j^2}\right) = \tan^{-1}\left(\frac{k}{k+1}\right)$. Then $\sum_{j=1}^{k+1}\tan^{-1}\left(\frac{1}{2j^2}\right) = \tan^{-1}\left(\frac{k}{k+1}\right) + \tan^{-1}\left(\frac{1}{2(k+1)^2}\right)$. Using the addition formula with $a = \frac{k}{k+1}$ and $b = \frac{1}{2(k+1)^2}$: $\frac{a+b}{1-ab} = \frac{\frac{k}{k+1}+\frac{1}{2(k+1)^2}}{1-\frac{k}{2(k+1)^3}} = \frac{2k(k+1)+1}{2(k+1)^2-k} = \frac{2k^2+2k+1}{2k^2+4k+2-k} = \frac{2k^2+2k+1}{2k^2+3k+2} = \frac{(k+1)(2k+1)}{(k+2)(2k+1)} = \frac{k+1}{k+2}$.

## Problem 3.30: Fermat Number Products and Series

Let $F_n = 2^{2^n} + 1$ be the $n$-th Fermat number for $n \geq 0$.

i) Show by mathematical induction that for any integer $k \geq 1$:

$$F_0F_1F_2\cdots F_{k-1} = F_k - 2$$

ii) Using the result from part (i), prove that for any integer $n \geq 1$:

$$1 - \sum_{k=0}^{n-1}\frac{1}{F_k} = \frac{2}{F_n - 1}$$

**Hint:**

i) • **Base Case:** Verify the identity for $k = 1$ by computing $F_0$ and $F_1$.

• **Inductive Step:** Assume the identity holds for $k = m$. Express $F_0F_1\cdots F_m$ using the assumption for $F_0F_1\cdots F_{m-1}$.

• Key insight: Use the difference-of-squares factorization $a^2 - 1 = (a-1)(a+1)$ with $a = 2^{2^m}$.

ii) • From part (i), $F_k - 1 = F_0F_1\cdots F_{k-1} + 1$ (rearranging the product identity).

• Consider the telescoping identity: $\frac{1}{F_k} = \frac{2}{F_{k-1}-1} - \frac{2}{F_k-1}$ for $k \geq 1$.

• Sum from $k = 1$ to $n - 1$ and combine with $\frac{1}{F_0}$ to form a telescoping series.

 

## Solution 3.30: Short

**Part i): Product Identity**

1. **Base Case ($k = 1$):** $F_0 = 2^{2^0} + 1 = 3$ and $F_1 - 2 = (2^{2^1} + 1) - 2 = 5 - 2 = 3$. Thus $F_0 = F_1 - 2$. ✓
2. **Inductive Step:** Assume $F_0F_1 \cdots F_{m-1} = F_m - 2$ for some $m \geq 1$. Consider the product for $k = m + 1$:

$$\begin{aligned} F_0F_1 \cdots F_{m-1}F_m &= (F_m - 2)F_m = (2^{2^m} + 1 - 2)(2^{2^m} + 1) \\ &= (2^{2^m} - 1)(2^{2^m} + 1) = (2^{2^m})^2 - 1 \\ &= 2^{2\cdot 2^m} - 1 = 2^{2^{m+1}} - 1 = (2^{2^{m+1}} + 1) - 2 = F_{m+1} - 2 \end{aligned}$$

   The identity holds for $k = m + 1$. By induction, it holds for all $k \geq 1$. ✓

**Part ii): Sum Identity**
From part (i), rearranging gives $F_k - 1 = F_0F_1 \cdots F_{k-1} + 1$.
Consider the telescoping relation for $k \geq 1$:

$$\frac{2}{F_{k-1} - 1} - \frac{2}{F_k - 1} = \frac{2(F_k - F_{k-1})}{(F_{k-1} - 1)(F_k - 1)}$$

Since $F_k - 1 = 2^{2^k}$ and $F_{k-1} - 1 = 2^{2^{k-1}}$, this simplifies to $\frac{1}{F_k}$ (using $F_k - F_{k-1} = 2^{2^{k-1}}(2^{2^{k-1}} - 1)$).
Thus $\frac{1}{F_k} = \frac{2}{F_{k-1}-1} - \frac{2}{F_k-1}$ for $k \geq 1$.
Summing from $k = 1$ to $n - 1$:

$$\sum_{k=1}^{n-1} \frac{1}{F_k} = \sum_{k=1}^{n-1} \left( \frac{2}{F_{k-1} - 1} - \frac{2}{F_k - 1} \right) = \frac{2}{F_0 - 1} - \frac{2}{F_{n-1} - 1} = 1 - \frac{2}{F_{n-1} - 1}$$

(since $F_0 = 3$ implies $F_0 - 1 = 2$).
Adding $\frac{1}{F_0} = \frac{1}{3}$ and adjusting:

$$\sum_{k=0}^{n-1} \frac{1}{F_k} = \frac{1}{3} + 1 - \frac{2}{F_{n-1} - 1} = 1 - \left( \frac{2}{F_{n-1} - 1} - \frac{1}{3} \right)$$

By careful manipulation using $F_n - 1 = 2^{2^n}$ and the product relation, this yields:

$$1 - \sum_{k=0}^{n-1} \frac{1}{F_k} = \frac{2}{F_n - 1} \quad ✓$$

## Takeaways 3.4

- **Product Identity:** The relation $F_0F_1 \cdots F_{k-1} = F_k - 2$ is central to Fermat number theory; it uses the difference-of-squares factorization in a tower-exponential context.
- **Induction with Algebra:** Combining mathematical induction with algebraic factorization is a powerful technique for exponential sequences.
- **Telescoping via Reciprocals:** The series identity demonstrates how product relations can yield telescoping sums when written as reciprocal differences.
- **Relative Primality:** This identity immediately implies $\gcd(F_i, F_j) = 1$ for $i \neq j$, showing why Fermat numbers are pairwise coprime.

 

## Problem 3.31: Advanced: Fibonacci Telescoping Series

Let $(F_n)_{n\geq 0}$ be the Fibonacci sequence defined by $F_0 = 0$, $F_1 = 1$, and $F_{n+2} = F_{n+1} + F_n$ for all $n \geq 0$.

1. Prove Catalan's identity: $F_n^2 - F_{n-r}\,F_{n+r} = (-1)^{n-r}F_r^2$ for integers $n \geq r \geq 0$.
2. Show that $\dfrac{3}{F_{2k-1}F_{2k+3}} = \dfrac{1}{F_{2k-1}F_{2k+1}} - \dfrac{1}{F_{2k+1}F_{2k+3}}$ for all integers $k \geq 1$.
3. Evaluate the infinite series $S = \displaystyle\sum_{k=1}^{\infty} \frac{1}{F_{2k-1}F_{2k+3}}$.

**Hint:**

- For (i), try induction on $n$ while holding $r$ fixed; use the recurrence to relate the $n$ and $n+1$ cases.
- For (ii), first derive $F_{n+4} - F_n = 3F_{n+2}$, then substitute $n = 2k-1$ to obtain $F_{2k+3} - F_{2k-1} = 3F_{2k+1}$.
- For (iii), use the identity from (ii) to write the summand as a difference $\frac{1}{3}(T_k - T_{k+1})$ and telescope.

## Solution 3.31: Short

1. Base case $n = r$: $F_r^2 - F_0F_{2r} = F_r^2 = (-1)^{r-r}F_r^2$. Inductive step follows by algebra with the recurrence, yielding $F_{n+1}^2 - F_{n+1-r}F_{n+1+r} = (-1)^{n+1-r}F_r^2$.
2. Using $F_{n+4} - F_n = 3F_{n+2}$ and setting $n = 2k-1$ gives $F_{2k+3} - F_{2k-1} = 3F_{2k+1}$. Hence
$$\frac{1}{F_{2k-1}F_{2k+1}} - \frac{1}{F_{2k+1}F_{2k+3}} = \frac{F_{2k+3} - F_{2k-1}}{F_{2k-1}F_{2k+1}F_{2k+3}} = \frac{3}{F_{2k-1}F_{2k+3}}.$$
3. Therefore
$$S = \sum_{k=1}^{\infty} \frac{1}{F_{2k-1}F_{2k+3}} = \frac{1}{3}\sum_{k=1}^{\infty}\Big(\frac{1}{F_{2k-1}F_{2k+1}} - \frac{1}{F_{2k+1}F_{2k+3}}\Big)$$
telescopes to $S = \frac{1}{3}\big(\frac{1}{F_1F_3}\big) = \frac{1}{3}\cdot\frac{1}{2} = \frac{1}{6}$.

## Takeaways 3.5

- **Telescoping pattern:** Casting terms as $c\,(T_k - T_{k+1})$ enables immediate cancellation.
- **Key identity:** $F_{n+4} - F_n = 3F_{n+2}$ underpins the simple difference form.
- **Catalan's identity:** Provides structural control over quadratic Fibonacci expressions.

 

## Problem 3.32: A Trigonometric Inequality via Induction

Let $\theta \in \left(0, \frac{\pi}{2}\right)$ and $n$ be a positive integer greater than 1. Prove that:

$$\left(\frac{1}{\sin^n \theta} - 1\right)\left(\frac{1}{\cos^n \theta} - 1\right) \geqslant 2^n - 2^{\frac{n}{2}+1} + 1$$

**Hint:**

- Proceed with mathematical induction on $n$. First, verify the base case for $n = 2$.
- For the inductive step, express the $(n+1)$ case in a way that allows you to substitute your inductive hypothesis. Factoring out $\frac{1}{\sin\theta\cos\theta}$ from the expanded product is a good starting point.
- You will need to establish a lower bound for the remaining terms in your expansion. The AM-GM inequality will be useful here.
- To simplify and bound the resulting symmetric trigonometric expressions, use the substitution $t = \sin\theta + \cos\theta$. Remember to determine the valid range for $t$ given $\theta \in \left(0, \frac{\pi}{2}\right)$.

## Solution 3.32: Brief Solution

We proceed by mathematical induction. For the base case $n = 2$, evaluating both sides yields 1, so the proposition holds.
Assume the inequality is true for some integer $n \geqslant 2$. For the $n+1$ case, we can expand and algebraically manipulate the left-hand side into the following form:

$$\frac{1}{\sin\theta\cos\theta}\left[\left(\frac{1}{\sin^n\theta} - 1\right)\left(\frac{1}{\cos^n\theta} - 1\right) + \frac{1-\cos\theta}{\sin^n\theta} + \frac{1-\sin\theta}{\cos^n\theta} - 1\right] + 1$$

By applying the inductive hypothesis to the first part of the bracketed expression, and the AM-GM inequality to the middle terms, we establish a lower bound that relies on evaluating $\sqrt{\frac{(1-\cos\theta)(1-\sin\theta)}{\sin^n\theta\cos^n\theta}}$.
Using the property $\sin\theta\cos\theta \leqslant \frac{1}{2}$ and applying the substitution $t = \sin\theta + \cos\theta$ (where $t \in (1, \sqrt{2}]$), we can simplify the numerator and bound this square root term from below by $2^{\frac{n}{2}} - 2^{\frac{n-1}{2}}$.
Substituting these bounds back into our main expression cascades into the exact required form:

$$2^{n+1} - 2^{\frac{n+1}{2}+1} + 1$$

This completes the inductive step, proving the proposition is true for all $n \geqslant 2$.

 

**Takeaways 3.6**

- **Induction beyond sequences:** Mathematical induction is a highly effective tool for establishing bounds in parameterized inequalities, not just for proving integer sequence formulas.
- **Bridging steps with classical inequalities:** Combining induction with the AM-GM inequality often bridges the critical algebraic gap between the $n$ and $n+1$ steps in complex proofs.
- **The power of substitution:** When faced with symmetric trigonometric expressions, the substitution $t = \sin\theta + \cos\theta$ is a powerful technique to translate trigonometric bounds into manageable algebraic polynomials.

**Problem 3.33: Strengthening the Hypothesis**

For any positive integer $n$, prove that:

$$\frac{1}{2} \cdot \frac{3}{4} \cdot \dots \cdot \frac{2n-1}{2n} < \frac{1}{\sqrt{3n}}$$

**Hint:**

- Try applying standard mathematical (weak) induction directly. What happens when you try to prove that the inductive step implies the $(n+1)$ case?
- The direct algebraic manipulation breaks down because the hypothesis isn't quite strong enough to bridge the gap to the next step.
- Try proving a *stronger* (tighter) bound. Can you replace the right side of the inequality with something slightly smaller, such as $\frac{1}{\sqrt{3n+1}}$, and apply induction to that new statement instead?

 

## Solution 3.33: Brief Solution

If we attempt to prove the inequality directly using induction, the inductive step requires us to show that $\frac{1}{\sqrt{3n}} \cdot \frac{2n+1}{2(n+1)} \leqslant \frac{1}{\sqrt{3(n+1)}}$. However, algebraically, this simplifies to $(n+1)(2n+1)^2 \leqslant n(2n+2)^2$, which is false.

Instead, we prove a strengthened proposition:

$$\frac{1}{2} \cdot \frac{3}{4} \cdot \cdots \cdot \frac{2n-1}{2n} \leqslant \frac{1}{\sqrt{3n+1}}$$

- **Base Case ($n = 1$):** The LHS is $\frac{1}{2}$, and the RHS is $\frac{1}{\sqrt{3(1)+1}} = \frac{1}{2}$. Thus, $\frac{1}{2} \leqslant \frac{1}{2}$ holds true.
- **Inductive Step:** Suppose the strengthened proposition is true for some positive integer $k$. For $k+1$, we must prove:
$$\frac{1}{\sqrt{3k+1}} \cdot \frac{2k+1}{2k+2} \leqslant \frac{1}{\sqrt{3(k+1)+1}} = \frac{1}{\sqrt{3k+4}}$$
Squaring both sides to remove the radicals and cross-multiplying yields:
$$(3k+4)(2k+1)^2 \leqslant (3k+1)(2k+2)^2$$
Expanding both sides:
$$(3k+4)(4k^2+4k+1) \leqslant (3k+1)(4k^2+8k+4)$$
$$12k^3+28k^2+19k+4 \leqslant 12k^3+28k^2+20k+4$$
Subtracting common terms from both sides leaves:
$$0 \leqslant k$$
Since $k$ is a positive integer ($k \geqslant 1$), this inequality is always true. Thus, by mathematical induction, the strengthened proposition holds for all $n \geqslant 1$.

Finally, because $\frac{1}{\sqrt{3n+1}} < \frac{1}{\sqrt{3n}}$ for all positive integers $n$, it logically follows that the original strict inequality is true.

## Takeaways 3.7

- **The Paradox of Strengthening:** Sometimes it is difficult or impossible to realize the inductive step for a proposition $P(n)$. However, proving a *stronger* proposition $Q(n)$ (where $Q(n) \implies P(n)$) can actually be *easier*.
- **Why it works:** By strengthening the proposition, you are demanding more of what you need to prove, but you also give yourself a much stronger starting assumption (the inductive hypothesis) to use during the algebraic heavy lifting of the inductive step.

## Problem 3.34: Forward-Backward (Cauchy) Induction

(Jensen's Inequality) Suppose $f(x)$ is a convex function on $[a, b]$ (namely, for any $x, y \in [a, b]$, we always have $f\left(\frac{x+y}{2}\right) \leqslant \frac{1}{2}(f(x) + f(y))$).

Prove by induction, or otherwise, that for any $n$ numbers $x_1, \ldots, x_n \in [a, b]$, it is always true that
$$f\left(\frac{x_1 + \cdots + x_n}{n}\right) \leqslant \frac{1}{n}(f(x_1) + \cdots + f(x_n)).$$

 

**Hint:**

- Standard mathematical induction ($n \implies n+1$) is very difficult to apply here. Instead, try jumping forward. Can you easily prove the proposition for powers of 2 (i.e., $n = 2, 4, 8, \dots$)?
- Once you establish that the inequality holds for all $n = 2^k$, you need a way to fill in the gaps. Can you prove that if the inequality holds for some integer $k$, it must also hold for $k-1$?
- For the "backward" step ($k \implies k-1$), try substituting the average of the first $k-1$ terms as your $k$-th term in the inequality.

## Solution 3.34: Brief Solution

We prove this using Cauchy's Forward-Backward Induction.
**Part 1: Forward Step (Prove for $n = 2^k$)** We use standard induction on $k$.

- **Base Case ($k = 1, n = 2$):** This is given directly by the definition of convexity: $f\left(\frac{x_1+x_2}{2}\right) \leqslant \frac{f(x_1)+f(x_2)}{2}$.
- **Inductive Step:** Assume true for $n = 2^k$. For $n = 2^{k+1}$, we can group the terms into two halves of size $2^k$:

$$f\left(\frac{x_1 + \cdots + x_{2^{k+1}}}{2^{k+1}}\right) = f\left(\frac{\frac{x_1+\cdots+x_{2^k}}{2^k} + \frac{x_{2^k+1}+\cdots+x_{2^{k+1}}}{2^k}}{2}\right)$$

By the base case, this is $\leqslant \frac{1}{2}\left[f\left(\frac{x_1+\cdots+x_{2^k}}{2^k}\right) + f\left(\frac{x_{2^k+1}+\cdots+x_{2^{k+1}}}{2^k}\right)\right]$. Applying the inductive assumption to each half yields $\leqslant \frac{1}{2^{k+1}}(f(x_1) + \cdots + f(x_{2^{k+1}}))$. Thus, it holds for all powers of 2.

**Part 2: Backward Step (Prove $n = k \implies n = k-1$)** Assume the inequality holds for $k$ terms. We wish to show it holds for $k-1$ terms. Let $A = \frac{x_1+\cdots+x_{k-1}}{k-1}$ be the average of the first $k-1$ terms. We apply our $k$-term assumption to the set $\{x_1, \dots, x_{k-1}, A\}$:

$$f\left(\frac{x_1 + \cdots + x_{k-1} + A}{k}\right) \leqslant \frac{f(x_1) + \cdots + f(x_{k-1}) + f(A)}{k}$$

Notice that the left side simplifies: $\frac{x_1+\cdots+x_{k-1}+A}{k} = \frac{(k-1)A+A}{k} = A$. So the inequality becomes:

$$f(A) \leqslant \frac{f(x_1) + \cdots + f(x_{k-1}) + f(A)}{k}$$

Multiplying by $k$ and subtracting $f(A)$ from both sides gives:

$$(k-1)f(A) \leqslant f(x_1) + \cdots + f(x_{k-1})$$

Dividing by $k-1$ yields:

$$f(A) = f\left(\frac{x_1 + \cdots + x_{k-1}}{k-1}\right) \leqslant \frac{1}{k-1}(f(x_1) + \cdots + f(x_{k-1}))$$

This completes the backward step. Since we can reach any arbitrarily large power of 2, and step backwards to any integer below it, the inequality holds for all positive integers $n$.

 

**Takeaways 3.8**

- **Forward-Backward Induction:** When sequential induction fails, jumping to a sparse subset of integers (like powers of 2) and proving a "step-down" proposition is a highly effective, albeit unconventional, logical structure.
- **Jensen's Inequality:** This is a foundational proof in mathematical analysis. This generalized inequality forms the backbone for proving many other famous algebraic bounds, including the Arithmetic Mean-Geometric Mean (AM-GM) inequality.

**Problem 3.35: Cubic Mean–Arithmetic Mean Inequality**

Let $x_1, x_2, \ldots, x_n \in \mathbb{R}^+$ be positive real numbers.

**(i) Warm-Up ($n = 2$).** Prove that for any $a, b > 0$:

$$\frac{a^3+b^3}{2} \geq \left(\frac{a+b}{2}\right)^3$$

**(ii) Generalisation ($n$ variables).** Using mathematical induction (Cauchy's forward-backward induction), or otherwise, prove the **Cubic Mean–Arithmetic Mean (CM-AM)** inequality for all $n \geq 1$:

$$\sqrt[3]{\frac{x_1^3 + x_2^3 + \cdots + x_n^3}{n}} \geq \frac{x_1 + x_2 + \cdots + x_n}{n}$$

**(iii) Equality Condition.** State and justify the necessary and sufficient conditions under which equality holds.

**(iv) Application.** Let $a, b, c > 0$ satisfy $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = a + b + c$. Prove that

$$a^3 + b^3 + c^3 \geq 3,$$

and determine the values of $a, b, c$ for which equality is achieved.

**Hint:**

- **Hint for (i):** Compute the difference $\frac{a^3+b^3}{2} - \left(\frac{a+b}{2}\right)^3$ and factorize the numerator.
- **Hint for (ii):** Use Cauchy's forward-backward induction in two stages.
  1. *Forward step:* If the inequality holds for $k$ variables, prove it holds for $2k$ by grouping the $2k$ terms into two halves and applying the $n = 2$ base case.
  2. *Backward step:* If it holds for $k$ variables, prove it holds for $k-1$ by substituting the arithmetic mean of the first $k-1$ variables as the $k$-th variable.
- **Hint for (iii):** Trace the equality condition through the $n = 2$ base case.
- **Hint for (iv):** Apply AM-HM (or Cauchy-Schwarz) to the constraint to bound $a + b + c$ from below; then chain with CM-AM.

 

## Solution 3.35: Sketch

**Part (i).** Compute the difference:

$$\frac{a^3+b^3}{2} - \left(\frac{a+b}{2}\right)^3 = \frac{4(a^3+b^3)-(a+b)^3}{8} = \frac{3(a+b)(a-b)^2}{8} \geq 0,$$

since $a+b>0$ and $(a-b)^2 \geq 0$.

**Part (ii) – Forward-Backward Induction.**

*Forward step ($k \to 2k$).* Assume the inequality holds for $k$ variables. For $2k$ variables, group them into two equal halves with means $A$ and $B$:

$$\frac{1}{2k}\sum_{i=1}^{2k} x_i^3 = \frac{1}{2}\left[\frac{1}{k}\sum_{i=1}^{k} x_i^3 + \frac{1}{k}\sum_{i=k+1}^{2k} x_i^3\right] \geq \frac{A^3+B^3}{2} \geq \left(\frac{A+B}{2}\right)^3 = \left(\frac{1}{2k}\sum_{i=1}^{2k} x_i\right)^3.$$

So the claim holds for all powers of 2.

*Backward step ($k \to k-1$).* Assume the inequality holds for $k$ variables. Set $M = \frac{1}{k-1}\sum_{i=1}^{k-1} x_i$ and apply the $k$-variable case to $x_1, \ldots, x_{k-1}, M$:

$$\frac{x_1^3+\cdots+x_{k-1}^3+M^3}{k} \geq \left(\frac{(k-1)M+M}{k}\right)^3 = M^3.$$

Rearranging: $x_1^3+\cdots+x_{k-1}^3 \geq (k-1)M^3$, i.e. the inequality holds for $k-1$ variables. Since every integer can be reached from a larger power of 2 by stepping down, the result holds for all $n \geq 1$.

**Part (iii).** Equality in (i) requires $a=b$. Tracing through both inductive steps forces all variables to be equal. Equality holds **if and only if** $x_1 = x_2 = \cdots = x_n$.

**Part (iv).** By AM-HM (or Cauchy-Schwarz: $(a+b+c)\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right) \geq 9$):

$$a+b+c \geq \frac{9}{a+b+c} \implies (a+b+c)^2 \geq 9 \implies a+b+c \geq 3.$$

Applying CM-AM for $n=3$ and using this bound:

$$\sqrt[3]{\frac{a^3+b^3+c^3}{3}} \geq \frac{a+b+c}{3} \geq 1,$$

so $a^3+b^3+c^3 \geq 3$. Equality holds when $a=b=c=1$ (both AM-HM and CM-AM are tight).

## Takeaways 3.9

- **Cauchy's Forward-Backward Induction in inequalities:** Proving for powers of 2 first, then stepping backwards, is the canonical approach for symmetric inequalities when the standard $k \to k+1$ step is unclear.
- **The backward substitution trick:** Setting the extra variable equal to the average of the rest preserves the mean on the right side and cleanly removes the extra term on the left.
- **Inequality chaining:** Combining a foundational inequality (AM-HM) to bound a sum, then feeding that bound into CM-AM, is a powerful technique for application problems with mixed constraints.

### Problem 3.36: The Ladder of Powers

Let $x_1, x_2, \ldots, x_n$ be positive real numbers.

**(i)** Using the Cauchy-Schwarz inequality, prove the Arithmetic Mean – Quadratic Mean (AM-QM) inequality:

$$\frac{x_1 + x_2 + \cdots + x_n}{n} \le \sqrt{\frac{x_1^2 + x_2^2 + \cdots + x_n^2}{n}}$$

**(ii)** By applying the same argument to the sequence $x_1^2, \ldots, x_n^2$, show that:

$$\sqrt{\frac{x_1^2 + \cdots + x_n^2}{n}} \le \sqrt[4]{\frac{x_1^4 + \cdots + x_n^4}{n}}$$

**(iii)** Apply Cauchy-Schwarz once more to deduce:

$$\frac{x_1 + x_2 + \cdots + x_n}{n} \le \sqrt[8]{\frac{x_1^8 + x_2^8 + \cdots + x_n^8}{n}}$$

**(iv)** Given additionally that $x_1 x_2 \cdots x_n = 1$, prove that

$$x_1^8 + x_2^8 + \cdots + x_n^8 \ge n,$$

and determine exactly when equality holds.

**Hint:**

- **Hint for (i)–(iii):** Recall the Cauchy-Schwarz inequality: $(a_1 b_1 + \cdots + a_n b_n)^2 \le (a_1^2 + \cdots + a_n^2)(b_1^2 + \cdots + b_n^2)$. Try setting all $b_i = 1$ and climbing the ladder by applying the same trick three times.
- **Hint for (iv):** Recall AM-GM: $\frac{x_1 + \cdots + x_n}{n} \ge \sqrt[n]{x_1 \cdots x_n}$. How does the product condition give you a lower bound for the arithmetic mean? Then chain that into Part (iii).

 

## Solution 3.36: Sketch

**Part (i).** Apply Cauchy-Schwarz with $a_i = x_i$, $b_i = 1$:

$$\left(\sum x_i\right)^2 \le \left(\sum x_i^2\right) \cdot n.$$

Dividing by $n^2$ and taking the square root gives AM $\le$ QM.
**Part (ii).** Replace $x_i$ by $x_i^2$ in the same argument:

$$\left(\sum x_i^2\right)^2 \le n \sum x_i^4.$$

Divide by $n^2$, take the square root to get $\frac{\sum x_i^2}{n} \le \sqrt{\frac{\sum x_i^4}{n}}$, then take the square root once more.
**Part (iii).** Replace $x_i$ by $x_i^4$ and repeat:

$$\left(\sum x_i^4\right)^2 \le n \sum x_i^8.$$

Taking the 4th root gives $\sqrt[4]{\frac{\sum x_i^4}{n}} \le \sqrt[8]{\frac{\sum x_i^8}{n}}$. Chaining (i) $\le$ (ii) $\le$ (iii) yields the result.
**Alternative for (i)–(iii) via Jensen's Inequality.** Since $f(x) = x^8$ is strictly convex ($f''(x) = 56x^6 > 0$), Jensen's inequality gives immediately:

$$\left(\frac{\sum x_i}{n}\right)^8 \le \frac{\sum x_i^8}{n}.$$

Taking the 8th root solves Part (iii) in one line.
**Part (iv).** By AM-GM applied to $x_1, \dots, x_n$ with product 1:

$$\frac{x_1 + \cdots + x_n}{n} \ge \sqrt[n]{x_1 \cdots x_n} = 1.$$

Chaining with Part (iii) and raising to the 8th power:

$$1 \le \left(\frac{\sum x_i}{n}\right)^8 \le \frac{\sum x_i^8}{n} \implies \sum x_i^8 \ge n.$$

*Alternative:* Apply AM-GM directly to $x_1^8, \dots, x_n^8$:

$$\frac{\sum x_i^8}{n} \ge \sqrt[n]{\prod x_i^8} = \left(\prod x_i\right)^{8/n} = 1.$$

Equality in both Cauchy-Schwarz and AM-GM requires all variables to be equal; since their product is 1, equality holds if and only if $x_1 = x_2 = \cdots = x_n = 1$.

## Takeaways 3.10

- **The "dummy variable" trick:** Setting $b_i = 1$ in Cauchy-Schwarz is one of the most versatile techniques for converting a single sum into a power mean bound.
- **The Ladder of Means:** Each application of Cauchy-Schwarz climbs one rung: AM $\le$ QM $\le$ 4th-power mean $\le$ 8th-power mean. This is a specific instance of the Generalized Power Mean Inequality, where $M_p = \left(\frac{\sum x_i^p}{n}\right)^{1/p}$ is increasing in $p$.
- **Scaffolding vs. elegance:** The step-by-step Cauchy-Schwarz approach builds mechanical understanding; Jensen's inequality and direct AM-GM reveal the same result in one or two lines. Both perspectives are valuable.

 

## Problem 3.37: Discovering Exact Bounds

The Fibonacci sequence $\{F_n\}$ satisfies: $F_1 = F_2 = 1$, $F_{n+2} = F_{n+1} + F_n$. Prove by induction, or otherwise, that

$$\sum_{i=1}^{n} \frac{F_i}{2^i} < 2$$

**Hint:**

- **Hint 1 (Induction):** If you let $S_n = \sum_{i=1}^{n} \frac{F_i}{2^i}$, standard weak induction on the inequality $S_n < 2$ will fail because $S_k < 2$ does not give you a tight enough bound to prove $S_{k+1} < 2$. You need to strengthen the hypothesis.
- **Hint 2 (Induction):** Calculate the first few partial sums $(S_1, S_2, S_3, S_4)$ and calculate the exact difference $2 - S_n$. Look very closely at the numerators of these differences. Do you see a familiar sequence?
- **Hint 3 (Otherwise):** If you choose not to use induction, consider the infinite series $S = \sum_{i=1}^{\infty} \frac{F_i}{2^i}$. You can use the recursive definition $F_i = F_{i-1} + F_{i-2}$ to split the sum and create an algebraic equation in terms of $S$.

 

## Solution 3.37: Brief Solution

**Method 1: Strengthened Induction** Let $S_n = \sum_{i=1}^{n} \frac{F_i}{2^i}$. Calculating the first few terms reveals a pattern in the deficit from 2:

- $S_1 = \frac{1}{2} \implies 2 - S_1 = \frac{3}{2} = \frac{F_4}{2^1}$
- $S_2 = \frac{3}{4} \implies 2 - S_2 = \frac{5}{4} = \frac{F_5}{2^2}$
- $S_3 = 1 \implies 2 - S_3 = 1 = \frac{8}{2^3} = \frac{F_6}{2^3}$

We hypothesize a stronger exact equality: $S_n = 2 - \frac{F_{n+3}}{2^n}$. We prove this by induction. The base case ($n = 1$) holds as shown above. Assume true for some integer $k$: $S_k = 2 - \frac{F_{k+3}}{2^k}$. For $k + 1$:

$$S_{k+1} = S_k + \frac{F_{k+1}}{2^{k+1}} = 2 - \frac{F_{k+3}}{2^k} + \frac{F_{k+1}}{2^{k+1}}$$

Finding a common denominator for the fractions:

$$S_{k+1} = 2 - \frac{2F_{k+3} - F_{k+1}}{2^{k+1}}$$

Using the Fibonacci recurrence, $F_{k+3} = F_{k+2} + F_{k+1}$, we can rewrite the numerator:

$$2F_{k+3} - F_{k+1} = F_{k+3} + (F_{k+2} + F_{k+1}) - F_{k+1} = F_{k+3} + F_{k+2} = F_{k+4}$$

Thus, $S_{k+1} = 2 - \frac{F_{k+4}}{2^{k+1}}$, completing the induction. Since $F_{n+3} > 0$ for all $n \geqslant 1$, it strictly follows that $S_n = 2 - \frac{F_{n+3}}{2^n} < 2$.

**Method 2: Infinite Series (Otherwise)** Let $S = \sum_{i=1}^{\infty} \frac{F_i}{2^i}$. Assuming the series converges, we can expand it using $F_i = F_{i-1} + F_{i-2}$ for $i \geqslant 3$:

$$S = \frac{1}{2} + \frac{1}{4} + \sum_{i=3}^{\infty} \frac{F_{i-1} + F_{i-2}}{2^i}$$

$$S = \frac{3}{4} + \frac{1}{2} \sum_{i=3}^{\infty} \frac{F_{i-1}}{2^{i-1}} + \frac{1}{4} \sum_{i=3}^{\infty} \frac{F_{i-2}}{2^{i-2}}$$

Shifting the indices back yields:

$$S = \frac{3}{4} + \frac{1}{2}\left(S - \frac{F_1}{2}\right) + \frac{1}{4}(S)$$

$$S = \frac{3}{4} + \frac{1}{2}S - \frac{1}{4} + \frac{1}{4}S \implies S = \frac{1}{2} + \frac{3}{4}S$$

Solving for $S$ gives $\frac{1}{4}S = \frac{1}{2} \implies S = 2$. Since all terms in the sequence are strictly positive, any finite partial sum $S_n$ must be strictly less than the infinite sum. Therefore, $\sum_{i=1}^{n} \frac{F_i}{2^i} < 2$.

## Takeaways 3.11

- **Equality over Inequality:** It is often drastically easier to prove an exact equality by induction than an inequality, because inequalities throw away information that you might need for the algebraic manipulations in the next step.
- **Connecting Recurrences to Series:** The recursive structure of the Fibonacci sequence maps perfectly onto the shift-and-scale techniques used to evaluate infinite series.

 

**Problem 3.38: Hidden Recurrences**

Consider the Fibonacci sequence $F_n$, defined as $F_0 = 0$, $F_1 = 1$, $F_2 = 1$, and $F_{n+2} = F_{n+1} + F_n$ for $n \geqslant 0$.
Let $f(x) = \frac{1}{1+x}$.
(i) Find $f(f(x))$ and $f(f(f(x)))$.
(ii) Guess a general formula for $f^{(n)}(x)$ and express it in terms of $F_n$. Prove your hypothesis by induction, or otherwise.
(iii) Let $g_n(x) = x + f(x) + f^{(2)}(x) + \cdots + f^{(n)}(x)$. Prove that:

$$g_n(1) = \frac{F_1}{F_2} + \frac{F_2}{F_3} + \cdots + \frac{F_{n+1}}{F_{n+2}}$$

**Hint:**

- **Hint 1:** For part (i), simplify the compound fractions completely. Look closely at the coefficients of $x$ and the constant terms; map these directly to the first few terms of the Fibonacci sequence.
- **Hint 2:** For part (ii), formulate your hypothesis for $f^{(n)}(x)$ as a rational function using $F_n$, $F_{n-1}$, and $F_{n+1}$. Use standard mathematical induction, relying on the definition $F_{n+2} = F_{n+1} + F_n$ for the inductive step.
- **Hint 3:** For part (iii), you do not need to calculate the sum from scratch. Evaluate your general formula for $f^{(k)}(x)$ from part (ii) at $x = 1$. Notice how $F_k + F_{k-1}$ simplifies nicely in both the numerator and denominator.

## Solution 3.38: Brief Solution

**(i) Functional Iterations:**

- $f(f(x)) = \frac{1}{1+\frac{1}{1+x}} = \frac{1}{\frac{1+x+1}{1+x}} = \frac{1+x}{2+x}$
- $f(f(f(x))) = \frac{1}{1+\frac{1+x}{2+x}} = \frac{1}{\frac{2+x+1+x}{2+x}} = \frac{2+x}{3+2x}$

**(ii) Hypothesis and Proof:** Mapping the coefficients from our iterations to the Fibonacci sequence:

- $f^{(1)}(x) = \frac{1}{1+x} = \frac{F_1+F_0x}{F_2+F_1x}$
- $f^{(2)}(x) = \frac{1+x}{2+x} = \frac{F_2+F_1x}{F_3+F_2x}$
- $f^{(3)}(x) = \frac{2+x}{3+2x} = \frac{F_3+F_2x}{F_4+F_3x}$

**Hypothesis:** $f^{(n)}(x) = \frac{F_n+F_{n-1}x}{F_{n+1}+F_nx}$
**Proof by Induction:** The base cases are verified above. Assume the hypothesis holds for some positive integer $k$. We evaluate the $k+1$ iteration:

$$f^{(k+1)}(x) = f(f^{(k)}(x)) = \frac{1}{1+\frac{F_k+F_{k-1}x}{F_{k+1}+F_kx}}$$

Multiplying the numerator and denominator by $F_{k+1}+F_kx$:

$$f^{(k+1)}(x) = \frac{F_{k+1}+F_kx}{F_{k+1}+F_kx+F_k+F_{k-1}x}$$

Grouping the terms in the denominator and applying the Fibonacci recurrence ($F_{k+2} = F_{k+1}+F_k$):

$$f^{(k+1)}(x) = \frac{F_{k+1}+F_kx}{(F_{k+1}+F_k)+(F_k+F_{k-1})x} = \frac{F_{k+1}+F_kx}{F_{k+2}+F_{k+1}x}$$

By mathematical induction, the general formula holds for all positive integers $n$.
**(iii) Evaluating the Sum:** We are given $g_n(x) = x + f(x) + f^{(2)}(x) + \cdots + f^{(n)}(x)$. We must evaluate $g_n(1)$, which means evaluating $f^{(k)}(1)$ for each term from $k=0$ to $n$ (where $f^{(0)}(x) = x$).
Using our proven formula from part (ii):

$$f^{(k)}(1) = \frac{F_k+F_{k-1}(1)}{F_{k+1}+F_k(1)}$$

Because $F_k+F_{k-1} = F_{k+1}$ and $F_{k+1}+F_k = F_{k+2}$, this simplifies beautifully to:

$$f^{(k)}(1) = \frac{F_{k+1}}{F_{k+2}}$$

Now, we evaluate each term in the sum $g_n(1)$:

- $k=0: \quad f^{(0)}(1) = 1 = \frac{1}{1} = \frac{F_1}{F_2}$
- $k=1: \quad f^{(1)}(1) = \frac{1}{2} = \frac{F_2}{F_3}$
- $k=2: \quad f^{(2)}(1) = \frac{2}{3} = \frac{F_3}{F_4}$
- $\ldots$
- $k=n: \quad f^{(n)}(1) = \frac{F_{n+1}}{F_{n+2}}$

Summing these terms together directly yields the required proof:

$$g_n(1) = \frac{F_1}{F_2} + \frac{F_2}{F_3} + \cdots + \frac{F_{n+1}}{F_{n+2}}$$

 

**Takeaways 3.12**

- **Connecting Functions and Sequences:** Repeatedly composing a simple fractional transformation elegantly maps onto linear recurrence relations like the Fibonacci sequence.
- **Evaluation Simplifies Structure:** Substituting a specific value (like $x = 1$) into a generalized algebraic formula often collapses complex variables into simpler sequence relationships, making seemingly difficult series proofs straightforward.

**Problem 3.39: Recurrence Relation from Explicit Formula**

Consider the sequences $A_n$ and $B_n$ defined by

$$A_n = (1+\sqrt{2})^n + (1-\sqrt{2})^n$$
$$B_n = \sqrt{2}\left[(1+\sqrt{2})^n - (1-\sqrt{2})^n\right]$$

for integers $n \geq 0$.

**(i)** By finding a quadratic equation with roots $\alpha = 1+\sqrt{2}$ and $\beta = 1-\sqrt{2}$, show that both sequences satisfy the linear second-order recurrence relation $U_n = 2U_{n-1} + U_{n-2}$ for $n \geq 2$.

**(ii)** Using mathematical induction and the recurrence relation, prove that $A_n$ is an even integer for all integers $n \geq 0$.

**(iii)** Similarly, prove that $B_n$ is a multiple of 4 for all integers $n \geq 1$, and deduce that $B_n$ is an even integer when $n$ is an even positive integer.

**Hint:**

- **Part (i):** Calculate the sum of the roots $(\alpha+\beta)$ and the product of the roots $(\alpha\beta)$ to construct a characteristic quadratic equation of the form $x^2 - (\text{sum})x + (\text{product}) = 0$. Multiply this equation by $x^{n-2}$.
- **Part (ii):** Strong induction requires you to establish two base cases ($n = 0$ and $n = 1$) because the recurrence relation relies on the two previous terms.
- **Part (iii):** Calculate $B_1$ and $B_2$ for your base cases. In your inductive step, express the assumption as $B_k = 4P$ and $B_{k-1} = 4Q$ for integers $P$ and $Q$.

 

## Solution 3.39: Brief

**Part (i)** Let $\alpha = 1 + \sqrt{2}$ and $\beta = 1 - \sqrt{2}$. The sum of the roots is $\alpha + \beta = 2$. The product of the roots is $\alpha\beta = (1 + \sqrt{2})(1 - \sqrt{2}) = 1 - 2 = -1$. Therefore, $\alpha$ and $\beta$ are roots of the characteristic equation $x^2 - 2x - 1 = 0$. Since $\alpha$ is a root, $\alpha^2 = 2\alpha + 1$. Multiplying by $\alpha^{n-2}$ gives $\alpha^n = 2\alpha^{n-1} + \alpha^{n-2}$. By identical reasoning, $\beta^n = 2\beta^{n-1} + \beta^{n-2}$. Since both $A_n = \alpha^n + \beta^n$ and $B_n = \sqrt{2}(\alpha^n - \beta^n)$ are linear combinations of $\alpha^n$ and $\beta^n$, they must both satisfy the recurrence relation:

$$U_n = 2U_{n-1} + U_{n-2}$$

**Part (ii)** *Base cases:* For $n = 0$: $A_0 = (1+\sqrt{2})^0 + (1-\sqrt{2})^0 = 1 + 1 = 2$, which is an even integer. For $n = 1$: $A_1 = (1+\sqrt{2})^1 + (1-\sqrt{2})^1 = 2$, which is an even integer. *Inductive step:* Assume the statement is true for $n = k$ and $n = k - 1$ for some integer $k \geq 1$. That is, assume $A_k$ and $A_{k-1}$ are both even integers. From the recurrence relation established in (i):

$$A_{k+1} = 2A_k + A_{k-1}$$

Since $A_k$ is an integer, $2A_k$ is an even integer. Since $A_{k-1}$ is assumed to be an even integer, the sum of two even integers $(2A_k + A_{k-1})$ is also an even integer. Thus, $A_{k+1}$ is an even integer. By the principle of strong mathematical induction, $A_n$ is an even integer for all $n \geq 0$.

**Part (iii)** *Base cases:* For $n = 1$: $B_1 = \sqrt{2}\left[(1+\sqrt{2}) - (1-\sqrt{2})\right] = \sqrt{2}(2\sqrt{2}) = 4 = 4 \times 1$. For $n = 2$: $B_2 = \sqrt{2}\left[(1+\sqrt{2})^2 - (1-\sqrt{2})^2\right] = \sqrt{2}\left[(3+2\sqrt{2}) - (3-2\sqrt{2})\right] = \sqrt{2}(4\sqrt{2}) = 8 = 4 \times 2$. Both are multiples of 4. *Inductive step:* Assume true for $n = k$ and $n = k-1$ for some integer $k \geq 2$. Let $B_k = 4P$ and $B_{k-1} = 4Q$, where $P$ and $Q$ are integers. Using the recurrence relation:

$$B_{k+1} = 2B_k + B_{k-1}$$

$$B_{k+1} = 2(4P) + 4Q = 4(2P + Q)$$

Since $P$ and $Q$ are integers, $(2P+Q)$ is an integer. Thus, $B_{k+1}$ is a multiple of 4. By strong mathematical induction, $B_n$ is a multiple of 4 for all $n \geq 1$. Since any multiple of 4 is inherently an even number, $B_n$ is an even integer for all positive integers $n$, which inherently satisfies the condition for all *even* positive integers $n$.

## Takeaways 3.13

- **Binet's Formula Concept:** This problem introduces the inverse logic of Binet's formula (used for Fibonacci numbers). Instead of solving a recurrence relation to find an explicit surd formula, we use the explicit surd formula to find the much simpler integer-based recurrence relation.
- **Strong Induction:** Standard induction only requires the $k$th step to prove the $(k+1)$th step. Because second-order recurrence relations look back *two* steps, strong induction is required, necessitating two base cases and assumptions for both $k$ and $k - 1$.
- **Alternative Pathways:** Often in advanced mathematics, combinatorics (Binomial Theorem) and recursive sequences offer two distinct pathways to prove the exact same integer properties.

## Problem 3.40: The Paradox of Induction

Prove that for every positive integer $n$, the following inequality holds:

$$1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^n} < 2$$

 

**Hint:**

- **Hint 1:** Try proving this directly using standard induction. Notice that assuming the sum for $k$ is $< 2$ only allows you to prove the sum for $k+1$ is $< 2 + \frac{1}{2^{k+1}}$, which doesn't reach the required target of $< 2$.
- **Hint 2:** Consider the "Paradox of Induction": sometimes it is easier to prove a *stronger* statement because it gives you a more powerful inductive hypothesis.
- **Hint 3:** Instead of an inequality, try to find an exact formula for the series. Prove that $1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^n} = 2 - \frac{1}{2^n}$.

## Solution 3.40: Brief

Standard induction fails because the inductive step does not preserve the strict $< 2$ bound without extra information. We formulate a stronger proposition $Q(n)$:

$$1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^n} = 2 - \frac{1}{2^n}$$

**Base Case ($n = 1$):** LHS $= 1 + \frac{1}{2} = \frac{3}{2}$. RHS $= 2 - \frac{1}{2} = \frac{3}{2}$. $Q(1)$ is true.
**Inductive Step:** Assume $Q(k)$ is true: $1 + \cdots + \frac{1}{2^k} = 2 - \frac{1}{2^k}$.
For $n = k+1$, add $\frac{1}{2^{k+1}}$ to the sum:

$$\text{Sum} = \left(2 - \frac{1}{2^k}\right) + \frac{1}{2^{k+1}}$$

Make common denominators for the fractions:

$$\text{Sum} = 2 - \frac{2}{2^{k+1}} + \frac{1}{2^{k+1}} = 2 - \frac{1}{2^{k+1}}$$

Thus, $Q(k+1)$ is true. By induction, $Q(n)$ holds for all positive integers $n$.
Since $\frac{1}{2^n} > 0$ for all positive integers $n$, it follows directly that $2 - \frac{1}{2^n} < 2$, proving the original statement.

## Takeaways 3.14

- **The Paradox of Induction:** Counter-intuitively, a harder (stronger or more specific) theorem can be easier to prove by induction because the inductive hypothesis you get to assume is equally stronger.
- **Exact formulas beat loose bounds:** If an inequality induction gets stuck, look for an underlying equality that governs the pattern.

## Problem 3.41: Strong Induction with Rational Functions

Show that if $x + \frac{1}{x} \in \mathbb{Z}$, then $x^n + \frac{1}{x^n} \in \mathbb{Z}$ for all positive integers $n$.

 

**Hint:**

- **Hint 1:** This problem requires **Strong Induction**. To prove the statement for $n+1$, you will need to rely on the assumption that it is true for *both* $n$ and $n-1$.
- **Hint 2:** You need two base cases: $n=1$ and $n=2$. Calculate $x^2+\frac{1}{x^2}$ by squaring $\left(x+\frac{1}{x}\right)$.
- **Hint 3:** For the inductive step, expand the product $\left(x+\frac{1}{x}\right)\left(x^k+\frac{1}{x^k}\right)$. How does this relate to $x^{k+1}+\frac{1}{x^{k+1}}$?

**Solution 3.41: Brief**

Let $P(n)$ be the statement "$x^n+\frac{1}{x^n}\in\mathbb{Z}$".
**Base Cases:**
$P(1)$ is true by the given hypothesis ($x+\frac{1}{x}\in\mathbb{Z}$).
For $P(2)$, we observe that $x^2+\frac{1}{x^2}=\left(x+\frac{1}{x}\right)^2-2$. Since $x+\frac{1}{x}$ is an integer, its square minus 2 is also an integer. Thus $P(2)$ holds.
**Inductive Step (Strong Induction):**
Assume $P(k)$ and $P(k-1)$ are true for some integer $k\geq 2$. We examine the term for $k+1$:

$$x^{k+1}+\frac{1}{x^{k+1}}=\left(x+\frac{1}{x}\right)\left(x^k+\frac{1}{x^k}\right)-\left(x^{k-1}+\frac{1}{x^{k-1}}\right)$$

We know:

1. $x+\frac{1}{x}\in\mathbb{Z}$ (hypothesis)
2. $x^k+\frac{1}{x^k}\in\mathbb{Z}$ (by $P(k)$)
3. $x^{k-1}+\frac{1}{x^{k-1}}\in\mathbb{Z}$ (by $P(k-1)$)

Because integers are closed under multiplication and subtraction, the entire expression $x^{k+1}+\frac{1}{x^{k+1}}$ must be an integer. Thus $P(k+1)$ is true. By the Strong Principle of Mathematical Induction, $P(n)$ is true for all positive integers $n$.

**Takeaways 3.15**

- **Identifying Strong Induction:** Whenever an algebraic sequence or manipulation links the $(k+1)$-th term to multiple preceding terms (not just the $k$-th term), Strong Induction is required.
- **Multiple Base Cases:** If your inductive step looks back two steps (to $k-1$), you must explicitly prove two consecutive base cases to get the chain started.

# 4 Conclusion

Induction is a core reasoning tool in the HSC Mathematics Extension 2 course. Mastery comes from repeated, reflective practice. Use these problems to sharpen the ability to hypothesize, algebraically manipulate, and communicate complete proofs. Best of luck with your studies and future competitions!

 

# 5 Appendix: Forms of Mathematical Induction

There are **two primary forms** of mathematical induction. While both forms are logically equivalent and rooted in the Peano axioms, they are structured differently to tackle different types of mathematical problems. The weak form is introduced in NESA syllabus while the strong form is included here as an enrichment.

## Quick Comparison

| Feature | The First Form ("Weak") | The Second Form ("Strong") |
|---|---|---|
| **Inductive Hypothesis** | Assume only the immediately preceding case is true. | Assume **all** preceding cases are true. |
| **Assumption Notation** | Assume $P(k)$ is true. | Assume $P(1), P(2), \dots, P(k)$ are true. |
| **Inductive Step** | Prove $P(k) \implies P(k+1)$ | Prove $[P(1) \wedge P(2) \wedge \cdots \wedge P(k)] \implies P(k+1)$ |
| **Problem Type** | The next term relies **only** on the previous term. | The next term relies on **multiple** previous terms. |

## 1. The First Form (Standard or "Weak" Induction)

The first form is the foundational form of induction. Conceptually, it is often compared to a line of dominos: proving the first domino falls, and that any falling domino knocks over the next one, guarantees the entire line falls.

To prove a proposition $P(n)$ for positive integers $n$:

- **Base Case:** Prove that $P(1)$ is true. *(Note: The base case can sometimes begin at $n = 0$, $n = 2$, or another integer depending on the domain of the proposition).*
- **Inductive Hypothesis:** Assume that $P(k)$ is true for some arbitrary positive integer $k$.
- **Inductive Step:** Using the assumption that $P(k)$ is true, prove that $P(k+1)$ is true.

**Applications:**
The first form is sufficient for the vast majority of standard proofs, including series summations, divisibility problems, calculus proofs (like general derivatives), and standard inequalities where the $(k+1)$th state relies *strictly* on the $k$th state.

## 2. The Second Form (Strong Induction)

The second form provides a broader assumption in the inductive step. Instead of assuming only the immediately preceding case is true, you assume *all* preceding cases are true.

To prove a proposition $P(n)$:

- **Base Case(s):** Prove that $P(1)$ is true. *(Note: Problems requiring strong induction often require proving multiple base cases, such as $P(1)$ and $P(2)$, to establish the foundation).*

- **Inductive Hypothesis:** Assume that $P(j)$ is true for all integers $j$ such that $1 \leq j \leq k$. This means assuming $P(1), P(2), \ldots, P(k)$ are all valid.
- **Inductive Step:** Prove that $P(k+1)$ is true, substituting in any or all of the previous cases $P(1)$ through $P(k)$ as needed to complete the algebra.

**Applications:**
Strong induction is highly useful when dealing with non-linear recursive sequences. If a sequence is defined such that a term depends on multiple previous terms (e.g., $x_n = 3x_{n-1} - 2x_{n-2}$), standard induction fails because knowing just $P(k)$ is not enough to find $P(k+1)$. You need the "strength" of knowing $P(k)$ and $P(k-1)$ simultaneously. It is also frequently used in number theory (e.g., proving the Fundamental Theorem of Arithmetic).

## Note on Logical Equivalence

It is worth emphasizing to students that the "Second Form" is not actually mathematically stronger than the "First Form". They are logically equivalent. If a proposition can be proved by strong induction, it can technically be proved by standard induction (by defining a new cumulative proposition $Q(n)$). However, utilizing the Second Form directly bypasses convoluted logical gymnastics, making it a much more elegant and practical tool for rigorous proof writing.

 

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